Distribution-Independent Tour-Length Estimation for the Traveling Salesman Problem Across Dimensions

Stephen Feldman, James P. Bailey, Bahar Çavdar
Rensselaer Polytechnic Institute, Troy, NY
{feldms5, bailej6, cavdab2}@rpi.edu

Abstract

We present the Geometrically Assisted Regression Tree (GART) 2.0, a distribution-independent estimator of Traveling Salesman Problem (TSP) tour length whose feature set is defined for arbitrary dimension. Existing estimators either assume a known sampling density or depend on planar geometry that becomes unstable as dimension grows. GART 2.0 is a LightGBM regressor that predicts the instance-specific ratio $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$ of tour length to minimum-spanning-tree (MST) length and returns $\hat{L}_{\mathrm{TSP}} = \hat{\alpha}\, L_{\mathrm{MST}}$; no distribution label, no density, and no convex hull enters the computation. Trained on synthetic instances alone, it transfers to TSPLIB95 and, through a multidimensional-scaling embedding, to non-Euclidean instances given only as a distance matrix. On all four benchmark aggregates GART 2.0 leads every baseline scored there, from a seven-baseline roster, and on the multidimensional benchmark its dispersion falls monotonically in the two arguments the feature set was built to carry—growing instance size and growing dimension—out to a hundred dimensions, a value withheld from training entirely. We then score a certified Held–Karp 1-tree lower bound, which consumes no training corpus and proves its own output, and use it to fix the boundary of the method. On the corpus aggregate the bound is the better instrument; that aggregate is mostly small instances, where the relaxation is nearly exact and nearly free, and read at fixed instance size the ordering turns over. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the plane it is strictly better than the bound on both axes from an ascent budget of 25 upward. The reason is structural: the bound pays a quadratic price in instance size for a fixed accuracy in every dimension, while the estimator's cost carries no accuracy term—so the estimator sharpens exactly where an exact solve becomes infeasible. Below that size, and off the training grid in dimension, the bound is preferable; both halves of that rule are needed to use either method well. The same bound doubles as an acceptance test for the labels every method is scored against: every label it reaches passes, with zero violations.

Keywords Traveling Salesman Problem · Tour-length estimation · Distribution-independent · Arbitrary dimension

1 Introduction

The Traveling Salesman Problem (TSP) appears throughout logistics operations planning, network design, and resource allocation [Chien, 1992, Applegate et al., 2006]. The TSP is NP-hard [Karp, 1972, Papadimitriou, 1977], so rapid tour-length estimates are essential for real-time decision-making. The TSP often enters as a subproblem of a larger optimization, and only its cost is needed. A facility-location decision, for example, requires the route cost of every candidate, so a full solve is paid once per candidate.

1.1 Background and Motivation

Last-mile delivery accounts for approximately 53% of total shipping costs by one industry estimate [Finmile], and the UPS ORION system is reported to save roughly 100 million miles and \$300–400 million annually through route optimization [Holland et al., 2017]. In such systems orders arrive continuously and routes are long, so the cost of a full solve is paid many times over. An estimator replaces that solve wherever only the relative cost of candidate routes matters.

The same estimation problem recurs wherever a TSP appears as a subproblem, and it rarely arrives in the plane. Radiation hybrid mapping orders genetic markers by a TSP over an abstract similarity metric [Agarwala et al., 2000]. Mobile-sink routing in wireless sensor networks contains a TSP through the data-collection locations [Yuan and Peng, 2010]. Large-scale routing methods partition an instance into sub-TSPs and compare candidate partitions [Mariescu-Istodor and Fränti, 2021], and Pan et al. [2023] learn a hierarchical policy that repeatedly selects node subsets and routes through them. Each of these planners needs the cheaper option predicted as cheaper, so the estimator must be accurate and rank-preserving.

None of those settings is planar. Most classical tour-length estimators, however, are built for low-dimensional Euclidean inputs and depend on geometric quantities—most often convex-hull area—that become unstable or intractable as dimension grows [Chazelle, 1993]. An estimator for these settings must be defined for arbitrary d without building a hull. The same requirement opens the non-Euclidean case. A distance matrix admits a Euclidean representation of bounded distortion once enough dimensions are allowed [Bourgain, 1985], so an estimator that stays accurate at those dimensions can consume the embedding. Dimension-scalability is the design goal of this paper.

1.2 Limitations of the Current Estimators

No published estimator covers high-dimensional or non-Euclidean instances. The Beardwood–Halton–Hammersley (BHH) theorem [Beardwood et al., 1959] is stated for arbitrary dimension, but it is asymptotic in n and its calibrating constants and density integral are not recoverable from an arbitrary finite point set; published constants exist for $d = 2$ [Johnson et al., 1996] and $d = 3$ [Percus and Martin, 1996], and above that only a large-d approximation [Bertsimas and van Ryzin, 1990]. The estimators of Chien [1992] are planar and depend on particular depot and covering-region definitions. A parallel line of work is tuned to the 2D Euclidean case: Daganzo [1984b] gives a strip-based construction, Kwon et al. [1995] fit regression and neural models on rectangular-region predictors, and Çavdar and Sokol [2015] propose a distribution-free estimator built from per-axis spread statistics about the covering rectangle. Space-filling heuristics such as the planar construction of Bartholdi and Platzman [1982] instead build a tour outright in $O(n \log n)$. Every member of this family consumes a convex-hull area or an axis-aligned covering box, and both degrade with dimension. The embedding route out of the non-Euclidean case therefore returns these estimators to the regime where their geometric inputs are unstable. Section 4 scores BHH and Çavdar and Sokol [2015] numerically, both on the full benchmarks and on the subset where their published assumptions hold (Table 16). We survey Chien [1992], Daganzo [1984b] and Kwon et al. [1995] without scoring them. We could not obtain their primary sources, and we do not publish numbers built on constants we have not read in the original.

Recent neural approaches address the same gap from the learning side. Varol et al. [2024] report sub-1% deviation on standard distributions by combining a multilayer perceptron (MLP) with domain-knowledge features, but their pipeline requires solution-level features, which are computationally expensive and challenging to generalize across all instance sizes. Kou et al. [2022] use the standard deviation of random feasible tour lengths as a single-feature, distribution-independent predictor on Euclidean and non-Euclidean instances, but the repeated tour sampling raises its per-instance cost. Kou et al. [2024] extend this principle beyond the TSP to the vehicle-routing (VRP), MST, and matching problems, again from the standard deviation of random feasible solutions. We exclude that sampling cost on the same grounds as solution-level features. The closest methodological neighbor is Kou et al. [2023], which pairs a Sammon-map embedding with a distribution-independent estimator on non-Euclidean TSPLIB95 instances. Our pipeline instead uses classical metric MDS for the embedding and computes MST-topology features on the original distance matrix rather than the embedded coordinates.

A separate classical line bounds the tour instead of estimating it. Held and Karp [1970, 1971] relax the Hamiltonian-cycle constraint to a *1-tree*, a spanning tree on all but one node plus the two cheapest edges at that node. Any vector of node penalties yields a proven lower bound on the optimal tour, so an ascent over penalties stopped anywhere returns a certificate. The bound needs no training corpus and no distributional assumption, so it has no split to contaminate and no domain to leave. It never invokes the triangle inequality, so it is defined wherever a symmetric distance matrix is, including the non-Euclidean instances Section 6 reaches only through an embedding. Its cost is quadratic in n per ascent iteration, so accuracy is bought continuously. Section 5.1 gives the relaxation and the two step schedules we score as first-class rows on every benchmark (Table 1). Because the bound is certified, cheap in the plane only at small n, and quadratic in n everywhere, it fixes where a learned estimator is worth running. Section 5 draws that line.

1.3 Contribution

We present the Geometrically Assisted Regression Tree (GART) 2.0, a LightGBM regressor [Ke et al., 2017] that predicts a per-instance scaling factor $\hat{\alpha}$ rather than applying a fixed constant. It is distribution-independent. No distribution label and no parametric density is supplied at inference, although accuracy varies by distribution type. That property is not by itself new—Çavdar and Sokol [2015] and Kou et al. [2022, 2023] share it—but those estimators remain tied to 2D coordinate geometry or to per-instance sampling. The contribution of GART 2.0 is to be distribution-independent *and* dimension-scalable at once: all 31 features are defined for arbitrary d through MST topology, centroid dispersion and coordinate ranges, no feature requires a convex hull, and the same feature set applies unchanged as both dimension and instance size grow. Section 3.1 gives the feature set and Section 3.4 validates it against SHAP attributions on the held-out split.

We benchmark GART 2.0 against seven baselines: two classical estimators, three constant multiples of L_{MST} of which one is calibrated on the training split, the GART 1.0 predecessor [Feldman, 2025], and a space-filling-curve tour construction. Both classical estimators are scored on the full benchmarks and again on the i.i.d. uniform subset where their published assumptions hold, BHH there receiving the exact sampling region its theorem names. GART 2.0 has the lowest aggregate mean absolute percentage error (MAPE) and standard deviation of percent error (SDPE) among the baselines scored on each of the four benchmarks, ahead of a calibrated per-cell lookup table on the multidimensional set, of Çavdar–Sokol on planar i.i.d. uniform instances, and of the asymptotic MST ratio on TSPLIB EUC_2D. Dispersion falls monotonically as instances grow, and again as the ambient dimension grows, holding at a dimension withheld from training entirely.

Section 5 then fixes the boundary of the method against the certified Held–Karp 1-tree bound, which is defined on every instance in this paper and on some that GART 2.0 declines. On the corpus aggregate the bound is the better instrument. Stratified by size as well as by dimension (Table 5), the ordering turns over above a few hundred nodes, and the mechanism is the complexity separation of Section 5.5. Section 5.6 states the resulting rule in full.

The bound is also an acceptance test for the stored labels. $B > L$ refutes a label with no tour entering the test, and every label the bound reaches passes (Appendix E).

2 Theoretical Foundations

This paper predicts the ratio $\alpha = L_{\text{TSP}}/L_{\text{MST}}$. This ratio varies from instance to instance. Let $\mathcal{G} = (V, E)$ be a complete undirected graph on the node set $V = \{1, \ldots, n\}$ with non-negative edge costs c_{ij}, typically the Euclidean distance between points in $\mathbb{R}^d$. Throughout, n is the number of nodes and d is the dimension of the coordinate space. The TSP asks for a minimum-cost Hamiltonian cycle. Any feasible tour is a 2-regular spanning subgraph [Dantzig et al., 1954, Applegate et al., 2006], of which the minimum spanning tree (MST) is the cheapest connected relaxation.

Prim's algorithm constructs one in $O(n^2)$ on a complete graph [Prim, 1957]. For nodes in $\mathbb{R}^d$ the pairwise distance computation scales as $O(n^2 d)$, linear in d, so the MST is a computationally stable scaffold for high-dimensional inference. It also brackets the tour. Removing the heaviest edge from any tour leaves a spanning tree, so $L_{\mathrm{MST}} \le L_{\mathrm{TSP}}$; the double-tree heuristic [Rosenkrantz et al., 1977] doubles the MST edges into an Eulerian multigraph and shortcuts repeated nodes to yield a feasible tour of length at most $2\,L_{\mathrm{MST}}$ on any instance obeying the triangle inequality. Together these give

$$L_{\mathrm{MST}} \le L_{\mathrm{TSP}} \le 2 \cdot L_{\mathrm{MST}}. \tag{1}$$

The upper constant cannot be reduced in general: for collinear point sets the optimal tour must traverse the line to its far end and return, so $L_{\mathrm{TSP}} = 2\,L_{\mathrm{MST}}$ exactly.

A separate and frequently conflated result concerns *algorithmic* approximation, not this intrinsic ratio. The Christofides–Serdyukov algorithm [Christofides, 1976, Serdyukov, 1978] augments the MST with a minimum-weight perfect matching (MWPM) on the odd-degree nodes and returns a tour within a factor $\frac{3}{2}$ of the *optimum*; Karlin et al. [2021] improve on that factor with a different construction, a randomized $\frac{3}{2} - \epsilon$ approximation. Both are guarantees on an algorithm's output relative to the optimal tour, and neither tightens the intrinsic ratio $L_{\mathrm{TSP}}/L_{\mathrm{MST}}$, which can still equal 2. The MWPM costs $O(n^3)$ [Edmonds, 1965, Gabow, 1990, Cook and Rohe, 1999], far above the $O(n^2)$ MST construction, so the Christofides refinement is impractical as a fast estimation subroutine.

For symmetric metric instances we therefore treat $\alpha \in [1, 2]$ as an intrinsic, instance-dependent quantity to be predicted. For uniformly distributed points the TSP constant itself varies with dimension [Percus and Martin, 1996], and no published MST constant covers every d, so we establish the behavior of the ratio empirically: the standard deviation of α over our training split falls monotonically from 0.1837 at $d = 2$ to 0.0762 at $d = 50$. The relative difference in objective value, and not the computational complexity of either problem, decreases as d grows, so a static estimator calibrated in the plane is biased in higher dimensions. GART 2.0 predicts α from the observed instance features instead.

3 Methodology: The Geometrically Assisted Regression Tree

GART 2.0 extends the estimator of Feldman [2025]. It predicts $\alpha = L_{\mathrm{TSP}}/L_{\mathrm{MST}}$ with a LightGBM ensemble and returns $\hat{L}_{\mathrm{TSP}} = \hat{\alpha} L_{\mathrm{MST}}$. We constrain the regression target to $\alpha \in [1, 2]$, the metric-TSP interval implied by Eq. (1). Two raw corpus values fall outside that interval, one training row at 0.975 and one validation row at 2.060, and target preparation clips them to the endpoints; an interior clip one millionth inside each endpoint keeps the logit target of Section 3.3 finite for those two rows.

3.1 Feature Selection

We extract 31 features from each instance: eleven geometric, scale and centroid descriptors, ten MST edge-weight statistics, nine MST topology and clustering proxies, and one constructive ratio.

3.1.1 Geometric and Centroid Descriptors

The MST statistics deliberately factor out scale and spread. The geometric group supplies them. MST edge weights and the degree sequence are invariant to rotation, translation and coordinate-system choice. The geometric descriptors, such as bounding hypervolume and centroid distance, generalize to arbitrary d without a convex hull. They carry the scale information of the BHH volume term [Beardwood et al., 1959] and adapt centroid statistics from Çavdar and Sokol [2015]. The dimension d is an explicit input. Before any geometric descriptor is evaluated we rotate the coordinates into their principal-axis frame [Jolliffe, 2002], which costs $O(nd^2 + d^3)$ and is dominated by dense MST construction whenever n is large relative to d; the rotation leaves the rotation-invariant features untouched and makes the axis-aligned bounding-box descriptors independent of the input orientation. Appendix I lists the geometric group.

3.1.2 MST Edge and Topological Features

Nineteen features summarize the MST. Ten describe the edge-weight distribution: mean, standard deviation, skewness, kurtosis, maximum, and the 10/25/50/75/90 quantiles. Nine describe topology, among them dominance, gaps, leaves, degrees, large edges, and the raw and normalized MST diameter. The MST's total length is not among them, because it is the multiplicand L_{MST} and withholding it keeps the learned quantity a scale-free ratio. The thirty-first feature, `greedy_nn_over_mst`, is the length of an exact greedy nearest-neighbor tour divided by L_{MST}. It is the only input GART 2.0 adds to the 30-feature block it inherits, and it is a constructed upper bound on the target: a greedy tour is a feasible tour, so its length is at least L_{TSP} and the ratio is at least α. The feature also gates coverage. On the non-Euclidean path of Section 6, where the metric is given but coordinates are not, the estimator returns a status row rather than a prediction for any instance whose ratio falls outside $[1.035, 2.209]$. That interval is the range the ratio spans over the full 106,272-row corpus, training, validation and test together, the constant the implemented gate applies; the training split alone spans the narrower $[1.0465, 2.1295]$. Definitions and computational costs are reported in Appendix C.

3.2 Dataset Generation

To train a general model, we built a synthetic dataset from four one-dimensional generators: Uniform, Normal (Gaussian), Clustered (k-means centers), and Correlated (autoregressive across dimensions). Each axis of each instance draws its generator independently, so configurations are anisotropic and topology varies by axis. Sixteen letter codes map onto these four functions. Eleven resolve to the uniform draw, two to normal, two to clustered, and one to correlated. Measured over all 2,444,256 axes in the corpus, 68.7% are uniform, 12.5% clustered, 12.5% normal, and 6.2% correlated. The multidimensional results below should be read against that mix.

The full feature corpus comprises 106,272 solved instances with node counts $n \in [5, 1000]$ and dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$ (17 training-visible dimension values), plus a held-out extrapolation bucket at $d = 100$. Instances are partitioned into training (69,768), validation (19,584), and test (16,920) via a stratified split on $(d, n, \text{grid size})$; all $d = 100$ instances are locked into the test partition so the model never sees that dimension during training or validation. The full training grid is given in Appendix B (Table 11); extrapolation to $d = 100$ is evaluated in Section 4.3.1.

For each instance, the generation pipeline ran Concorde [Applegate et al., 2006] and LKH-3 [Helsgaun, 2017] and retained the shorter tour. The canonical feature table stores the retained tour cost but not solver-specific costs or Concorde certificates. We therefore call these values reference tour costs and do not claim that every stored row is independently auditable as optimal. Solver-native integer quantization is not carried into the released labels: every reference cost is recomputed in float64 from the released coordinates. Instances with $n \leq 10$ are labeled by an exact Held–Karp dynamic program, a certificate; larger instances carry the float64 length of the retained tour, an upper bound on the optimum that is exact wherever the retained permutation is. Appendix E states the validation every label passes and the certification hierarchy.

3.3 Model Architecture and Training

The estimator for α is a gradient-boosted decision tree (GBDT) ensemble [Friedman, 2001], implemented in LightGBM [Ke et al., 2017]. Trees grow sequentially, each fit to the residual left by the trees before it. The final prediction $\hat{\alpha}$ is the sum of all tree outputs.

The learner uses squared-error regression on the logit-transformed target $z = \log\big((\alpha - 1)/(2 - \alpha)\big)$, and predictions are returned through its inverse $\alpha = 1 + (1 + e^{-z})^{-1}$. The bound $\alpha \in [1, 2]$ is therefore structural: the inverse transform cannot leave the open interval $(1, 2)$, so the collinear extreme of Eq. (1) is approached rather than attained; a clip to $[1, 2]$ remains in the released wrapper as a guard against non-finite inputs, and it is never binding. On the

16,920 multidimensional test predictions $\hat{\alpha}$ fell in $[1.029, 1.922]$, so the transform is not saturating at either end. The hyperparameters are inherited and frozen, so Section 4 compares a tuned field against an untuned entrant; an Optuna search [Akiba et al., 2019] against in-distribution validation moved multidimensional MAPE by 0.011 percentage points while costing 0.16 points of TSPLIB MAPE, and we rejected it (Appendix J). Early stopping used cost-level MAPE on the validation split. The model was fit on the training split only, with no train–validation refit, and the test split was scored once after that fit. The resulting ensemble contains 1,118 trees with 148 leaves per tree as the growth cap and 147.80 realized on average; its configuration is reported in Appendix J.

Two inputs carry non-increasing monotone constraints, n and d, encoding that α falls toward 1 as an instance grows or the ambient dimension rises. The constraints are enforced inside the tree builder, so a sum of such trees cannot increase along a ceteris-paribus sweep of either variable; a probe over 1,000 held-out instances accordingly returns 100% non-increasing sweeps on both axes with zero violations, on log-spaced grids of 24 points, d from 2 to 200 and n from 5 to 4,000, deliberately overshooting the synthetic evaluation range. The probe overwrites one column of a real held-out instance and holds the other thirty at that instance's measured values, so a swept row corresponds to no realizable instance. It measures the shape of the learned function in one argument and says nothing about accuracy at those n and d.

A constant multiple of L_{MST} is trivially non-increasing on both axes, so the informative control is the same fit with the constraints removed, which holds monotonicity on 8.4% of the dimension sweeps and 19.0% of the size sweeps. The constraint is therefore doing the work.

Feature extraction, not tree traversal, dominates inference cost. The trained model performs $K = 1{,}118$ traversals with empirical mean root-to-leaf depth $\bar{D} \approx 11.83$ (maximum 40). The ensemble performs approximately $K\bar{D} = 1.3 \times 10^4$ comparisons per prediction, independent of n. With a dense distance matrix, the pipeline costs $O(n^2 d + nd^2 + d^3)$, combining pairwise distances and MST construction with the PCA rotation. In the plane the Euclidean MST is a subgraph of the Delaunay triangulation [Shamos and Hoey, 1975], so it can be built in $O(n \log n)$. Sorting MST edges adds $O(n \log n)$, and the remaining topological statistics cost $O(n)$. The greedy nearest-neighbor tour behind `greedy_nn_over_mst` does not take that path: it scans a dense distance matrix in $O(n^2)$ for $n \leq 3000$ and switches to precomputed 16-neighbor candidate lists with an exact nearest-unvisited fallback above that threshold. Below $n = 3000$ the greedy tour, not the MST, is therefore the leading term of 2D extraction.

3.4 Feature Importance

We assess the 31-feature set with mean absolute SHAP magnitudes [Lundberg and Lee, 2017] against the released booster, on a 5,000-row sample of the 16,920-row held-out test split. The MST dominance ratio contributes 26.5% of the total magnitude and `greedy_nn_over_mst` 23.9%; node count and dimension jointly contribute 14.1%. The greedy ratio ranking second on its first appearance is the case for adding it. Appendix D reports the complete ranking.

4 Benchmarking

Each results table reports SDPE, MAPE, the median absolute percentage error (the |Median| column), and the mean signed percent error (MSPE). Rows that predict a multiple of L_{MST} also carry $R^2_\alpha = 1 - \sum_i (\hat{\alpha}_i - \alpha_i)^2 / \sum_i (\alpha_i - \bar{\alpha})^2$, with $\bar{\alpha}$ the mean of α over the reported bucket. SDPE is the primary metric:

$$\mathrm{SDPE} = 100 \cdot \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (e_i - \bar{e})^2}, \qquad \bar{e} = \frac{1}{N} \sum_{i=1}^{N} e_i.$$

Here $e_i = (\hat{L}_i - L_i)/L_i$ is the signed relative error and N is the number of scored instances; the leading factor of 100 expresses SDPE in percentage points. In the results tables, bold marks GART 2.0, the model proposed here; it does not mark the best value in a column, and several baselines beat it in individual cells. Each SDPE carries a 95% bootstrap confidence interval, suppressed where a group holds fewer than three instances. The intervals are computed per estimator and ignore the pairing, so overlapping intervals establish nothing on their own. Paired tests of the close comparisons are in Section 4.7.

4.1 Benchmark Roster

We compare GART 2.0 against seven baselines in four families: classical region-based estimators, constant-ratio references, the GART 1.0 predecessor, and a constructive heuristic. Every classical row receives the quantity its source defines. Daganzo, Chien, and Kwon–Golden–Wasil circulate only as mutually inconsistent transcriptions; they are surveyed in Section 1.2 and scored nowhere in this paper (Appendix F).

Classical region-based estimators. Each estimator is gated to the domain its source covers. The BHH form is $\hat{L} = \beta_d n^{(d-1)/d} \mathcal{V}^{1/d}$, where $\mathcal{V}$ is the measure of the sampling region [Beardwood et al., 1959] and $\beta_2 = 0.7124$ [Johnson et al., 1996]; for $d = 3$ we use $\beta_3 = 0.6979$ [Percus and Martin, 1996] and for $d \geq 4$ the conjectured large-d asymptotic $\beta_d \approx \sqrt{d/(2\pi e)}$ [Bertsimas and van Ryzin, 1990]. Çavdar and Sokol [2015] gives $\hat{L} = 2.791\sqrt{n\,\mathrm{cstdev}_x \mathrm{cstdev}_y} + 0.2669\sqrt{n\,\mathrm{stdev}_x \mathrm{stdev}_y A/(\bar{c}_x \bar{c}_y)}$, where A is the area of the rectangle covering the nodes, stdev_j is the standard deviation of the raw coordinates on axis j, and $\bar{c}_j$ and cstdev_j are the mean and standard deviation of the absolute distances from the nodes to that rectangle's midpoint line on axis j. The Çavdar–Sokol estimate is divided by the finite-n adjustment factor its source fits, held fixed outside the node range that fit covers. The Çavdar–Sokol estimator, both of its constant sets, and its adjustment factor come from Çavdar [2014].

Constant-ratio references. The $\alpha = 1$ floor returns L_{MST} itself: it is the certified lower bound of Eq. (1), the only constant defined at every d, and the reference R^2_α is measured against. The asymptotic ratio uses the published constants $\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}} \approx 1.1253$ [Johnson et al., 1996, Cortina-Borja and Robinson, 2000] and is defined at $d = 2$ only, the one constant here taken from the literature instead of fitted on our data. The calibrated row uses $\hat{\rho}(d, n)$, the mean of α within each dimension–size cell of the training split. It is the best constant obtainable given everything cheap known about an instance. The $\alpha = 1$ and asymptotic rows are one estimator at two values of c in $\hat{L} = cL_{\mathrm{MST}}$, so their dispersion is related exactly by $\mathrm{SDPE}(c) = c\,\mathrm{SDPE}(1)$; both are kept because that identity does not extend to MAPE.

Prior model. GART 1.0 [Feldman, 2025] is our prior model and the one learned estimator in the baseline count. It is quadratic in construction and planar only: it appears only on the 2D synthetic and TSPLIB EUC_2D benchmarks (Table 17). Its artifact-driven extractor uses a dense pairwise-distance matrix, so its effective time and memory cost is $O(n^2)$ rather than $O(n \log n)$, and it treats the lexicographically first unique coordinate as a depot-like reference for two features although no depot is defined in these datasets. It differs from GART 2.0 in data, features, and training, so the comparison isolates no single causal change.

Constructive heuristic. The Hilbert row builds a tour from a bare d-dimensional Hilbert ordering after independent per-axis min–max normalization to a 16-bit grid. It constructs a tour rather than estimating one, and the planar guarantee of Bartholdi and Platzman [1982] does not transfer to this multidimensional variant.

Certified lower bound. The Held–Karp 1-tree bound [Held and Karp, 1970, 1971] returns a proven lower bound on the optimal tour, at a cost fixed by an explicit ascent budget k. Two rows carry it, differing only in the step rule of the subgradient ascent: the Volgenant–Jonker schedule [Volgenant and Jonker, 1982], which consults no upper bound, and the Polyak step [Polyak, 1969], which sizes each move from the gap to a constructively built tour. Both are scored on all four benchmarks – the 2D diverse set, the multidimensional set, TSPLIB EUC_2D, and the non-EUC_2D TSPLIB instances – both raw and scaled by a single constant fitted per budget on the training split, and Section 5 reports them in full.

Region measure and matched domains. A *matched domain* is the subset of a benchmark on which a classical estimator's own published assumptions hold, so that scoring it there measures the method rather than the mismatch. Every synthetic generator draws coordinates inside $[0, G]^d$ for a coordinate-grid side length G, so the sampling region measure is G^d exactly. BHH is the only estimator here whose source names that measure; on the synthetic benchmarks it appears as two rows, one given G^d and one given the convex hull of the realized sample. Both are reported because the region measure is the source-faithful input only where the generator is uniform on that region: a near-collinear point set has one-dimensional effective support, and handing an area-based estimator G^2 inflates its prediction by an amount that describes the data and not the estimator. TSPLIB defines no sampling region at all, so there BHH receives the convex hull as a stated plug-in. Çavdar–Sokol takes no region input: its A is the area of the rectangle covering the nodes, a statistic of the sample, so it has a single row on every benchmark and supplying it G^2 would be our construction rather than its authors'.

Table 1: Benchmark roster. "Domain" is the range each source fits or derives; Section 4.4 rescores the classical estimators inside it.

Estimator	**Formula**	**Domain**	**Cost**	**Source**
Classical region-based estimators				
BHH	$\beta_d\, n^{(d-1)/d} \mathcal{V}^{1/d}$, $\mathcal{V}$ the sampling-region measure	i.i.d. uniform, $n \to \infty$	$O(nd)$	Beardwood et al. [1959], Johnson et al. [1996], Bertsimas and van Ryzin [1990]
Çavdar–Sokol	$2.791\sqrt{n\,\mathrm{cstdev}_x \mathrm{cstdev}_y} + 0.2669\sqrt{n\,\mathrm{stdev}_x \mathrm{stdev}_y A/(\bar{c}_x \bar{c}_y)}$	Minimum-area rectangle; $n \geq 100$ correction	$O(nd)$	Çavdar [2014]
Constant-ratio references				
L_{MST} ($\alpha = 1$)	L_{MST}	All	$O(n^2 d)$	Lower bound of Eq. (1)
Asymptotic MST ratio	$(\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}})\, L_{\mathrm{MST}} \approx 1.1253\, L_{\mathrm{MST}}$	$d = 2$, uniform, $n \to \infty$	$O(n^2 d)$	Johnson et al. [1996], Cortina-Borja and Robinson [2000]
Calibrated MST ratio $\hat{\rho}(d, n)$	$\hat{\rho}(d, n)\, L_{\mathrm{MST}}$, per dimension–size cell	Trained cells	$O(n^2 d)$	This work
Prior model and constructive heuristic				
GART 1.0	2D-only feature set, GBDT regression on α	$d = 2$	$O(n^2)$	Feldman [2025]
Custom Hilbert sort	Bare d-dimensional ordering, per-axis normalized, Hilbert order $h = 16$	Constructs a tour	$O(n \log n)$	After Bartholdi and Platzman [1982]
Certified lower bound				
Held–Karp 1-tree, Volgenant–Jonker step, budget k	the incumbent $\hat{w}_k = \max_{j \leq k} w(\pi_j)$ over k ascent steps; step schedule consults no upper bound	Any symmetric distances, any d; certificate	$\Theta(n^2 d + kn^2)$	Held and Karp [1970, 1971], Volgenant and Jonker [1982], Helsgaun [2000]

Table 1 continued from the previous page

Estimator	Formula	Domain	Cost	Source
Held–Karp 1-tree, Polyak step, budget k	The same bound; step $\gamma(\mathrm{UB} - w)/\|g\|^2$ against a constructive tour	Any symmetric distances, any d; certificate	$\Theta(n^2 d + kn^2)$	Held and Karp [1970], Polyak [1969]
Calibrated 1-tree	c_k times either bound, c_k fitted per budget on the training split	As trained	$\Theta(n^2 d + kn^2)$	This work

4.2 Datasets

The evaluation covers a multidimensional synthetic set, a diverse 2D synthetic set, and TSPLIB95, whose EUC_2D instances form the third benchmark and whose screened non-EUC_2D instances form the fourth, constituted in Section 6. The synthetic sets are held-out splits from our generation pipeline (Appendix B). The evaluation assumes symmetric metric distances; Section 6 lists the ten EXPLICIT TSPLIB matrices whose structural triangle-inequality violations put them outside every metric-dependent aggregate here.

4.2.1 Multidimensional Synthetic

The benchmark contains 16,920 held-out instances with $n \in [5, 1000]$ and d drawn from the 17 training-visible values plus $d = 100$, sampled by stratified dimension–size–grid buckets from the same four per-axis generators as the training set: uniform, normal, clustered (k-means [Lloyd, 1982]), and autoregressive-correlated. Stratification enforces zero row overlap with the training split, but the generator families and axis mix of Section 3.2 are shared, so this benchmark measures held-out sampling and not held-out geometry. Section 4.5 withholds whole dimensions and whole size ranges from training to separate the two. The full (d, n) count grid is given in Appendix B.3 (Table 10).

4.2.2 2D Diverse Synthetic

The benchmark contains 2,580 instances with $d = 2$, $n \in [5, 1000]$, and coordinate-grid side length $G \in [1000, 10000]$. It was generated by the separate 2D benchmark pipeline; an identifier-level audit found zero overlap with the 106,272-row training/validation/test corpus. Five distribution classes (Table 2) each probe a different assumption made by classical 2D estimators. Two of the thirteen sub-generators test geometry the training corpus does not contain: Line Noise, which is near-collinear, and `grid`, a jittered square lattice. Appendix B.2 gives the construction of both classes.

Table 2: 2D benchmark dataset composition (2,580 instances).

Class	Generators	Count	Stress target
Isotropic [Çavdar and Sokol, 2015, Golden and Stewart, 1985]	Uniform, Normal, Triangular, Truncated Exp.	840	Control: $A_{\text{hull}} \approx A_{\text{box}}$.
Biased / Skewed [Çavdar and Sokol, 2015]	Squeezed, Uni-Tri, Tri-Squeezed, Correlated	840	Aspect-ratio distortion: $\sigma_x \neq \sigma_y$.
Geometric Struct. [Reinelt, 1991, Bossek et al., 2019]	Grid†, Boundary (hollow), X-Central	630	Lattice regularity, hollow voids.
Clustered [Golden and Stewart, 1985, Bossek et al., 2019]	k-Means ($k \in [5, 20]$)	60	Bimodal edge-weight distribution.
Line Noise†	Boundary-clipped $y = mx + b + \epsilon$; realized as a 2–4 segment union	210	Locally 1D structure; hull/box divergence.
Total		**2,580**	

† Not represented in training; included to measure generalization to unseen geometries.
`Grid` is one of the three Geometric Struct. generators; Section 4.3.2 scores it separately.

4.2.3 TSPLIB95 EUC_2D

The benchmark contains 78 EUC_2D instances with n from 51 (`eil51`) to 18,512 (`d18512`) [Reinelt, 1991]. Every numerically evaluated 2D baseline is defined on all 78, the paper's common instance set. The 23 retained non-EUC_2D instances are evaluated separately in Section 6.

4.3 Results

Each estimator in a bucket uses the stated common instance set.

All timings are descriptive single-machine measurements, and none supports an inferential comparison. TSPLIB reference-generation timing is omitted because the available logs have incomplete instance coverage and mix hardware sources. Two timing protocols appear in this paper, and no comparison mixes them. The Time columns of the multidimensional and 2D detail tables carry one cold harness invocation per instance, including interpreter and just-in-time warm-up. Table 17 and every ladder of Section 5 carry a solo protocol, warm-up outside the clock, run in a quiet window except where Section 5.4 discloses otherwise. The cold protocol reads an order of magnitude above the solo protocol for the same work. On the solo protocol, *corpus median* always means the median over the corpus of each instance's own median over its repeats.

4.3.1 Multidimensional Benchmark

The multidimensional benchmark is 16,920 held-out instances, of which the 16,846 with a recoverable reference cost are scored (Appendix E).

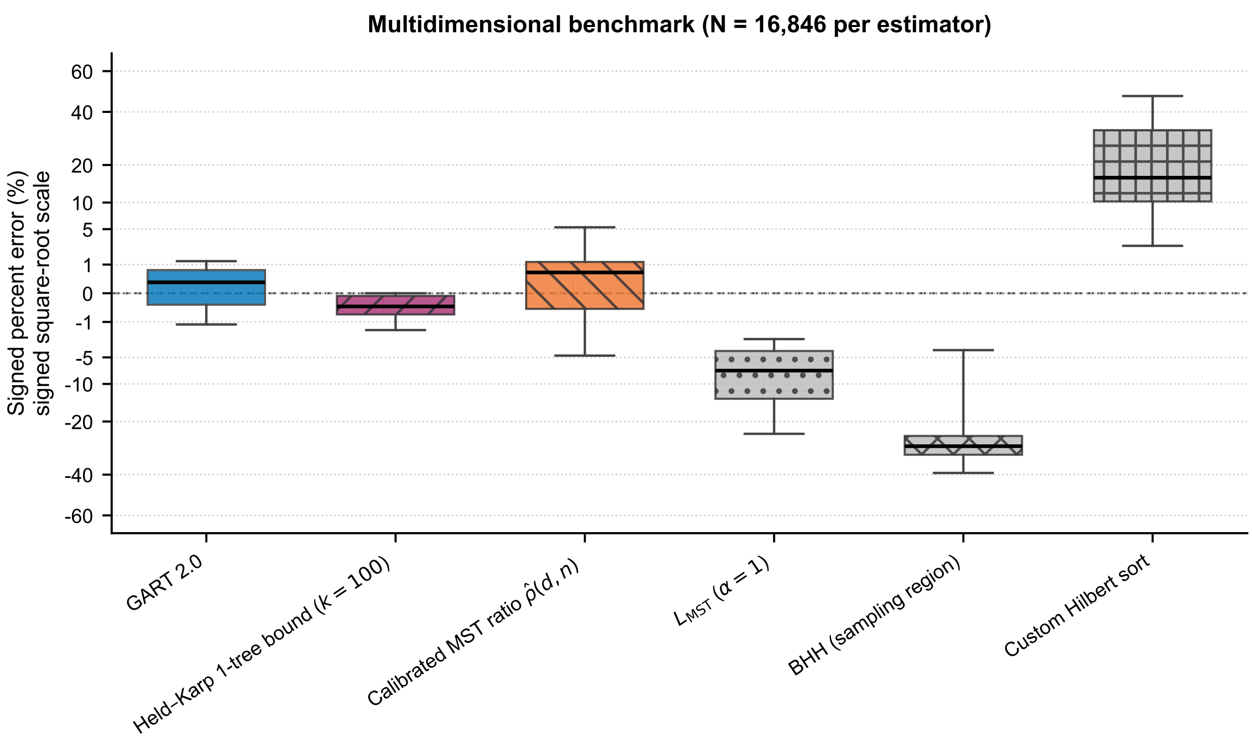

Figure 1: Signed percent error on the multidimensional benchmark ($N = 16{,}846$ per estimator); the certified 1-tree bound is drawn at ascent budget $k = 100$.

GART 2.0 obtains 0.62% MAPE and 0.99% SDPE overall. The strongest baseline in the roster of Section 4.1 that uses no learned model is the calibrated ratio $\hat{\rho}(d, n)$ at 1.82%/2.94%, so the learned α improves on a lookup table of

102 constants fitted on the same training split by a factor of 2.9 on MAPE and 3.0 on SDPE. The $\alpha = 1$ floor reaches 9.71%/7.26%, the custom Hilbert sort 21.21%/14.40%, and BHH given the exact sampling region 28.01%/13.57%. Figure 1 draws the certified Held–Karp 1-tree bound beside them. Its box lies wholly at or below zero because the bound cannot exceed the optimum. The sign of its error is therefore known before the instance is seen. Several estimator boxes also fall on one side of the line, but only as a fitted tendency observed after the fact. Section 5.3 scores that bound against GART 2.0 on cost and accuracy together.

SDPE decreases monotonically across all six size buckets, from 1.56% at $n \leq 10$ to 0.46% at $n \in [201, 500]$ and 0.45% at $n \in [501, 1000]$. Across dimension groups it falls from 2.37% at $d = 2$ to 0.48% at $d \in [30, 50]$, and falls again, to 0.44%, at the unseen $d = 100$, while MAPE rises there from 0.29% to 0.59%. The standard deviation of α over the training split falls monotonically with dimension, from 0.1837 at $d = 2$ to 0.0762 at $d = 50$, so the target concentrates as d grows and the prediction problem becomes easier in exactly the direction the extrapolation runs. The $d = 100$ result is therefore favorable-direction extrapolation, and Section 4.5 supplies the unfavorable-direction tests. Figure 2 draws the accuracy surface per dimension and size band.

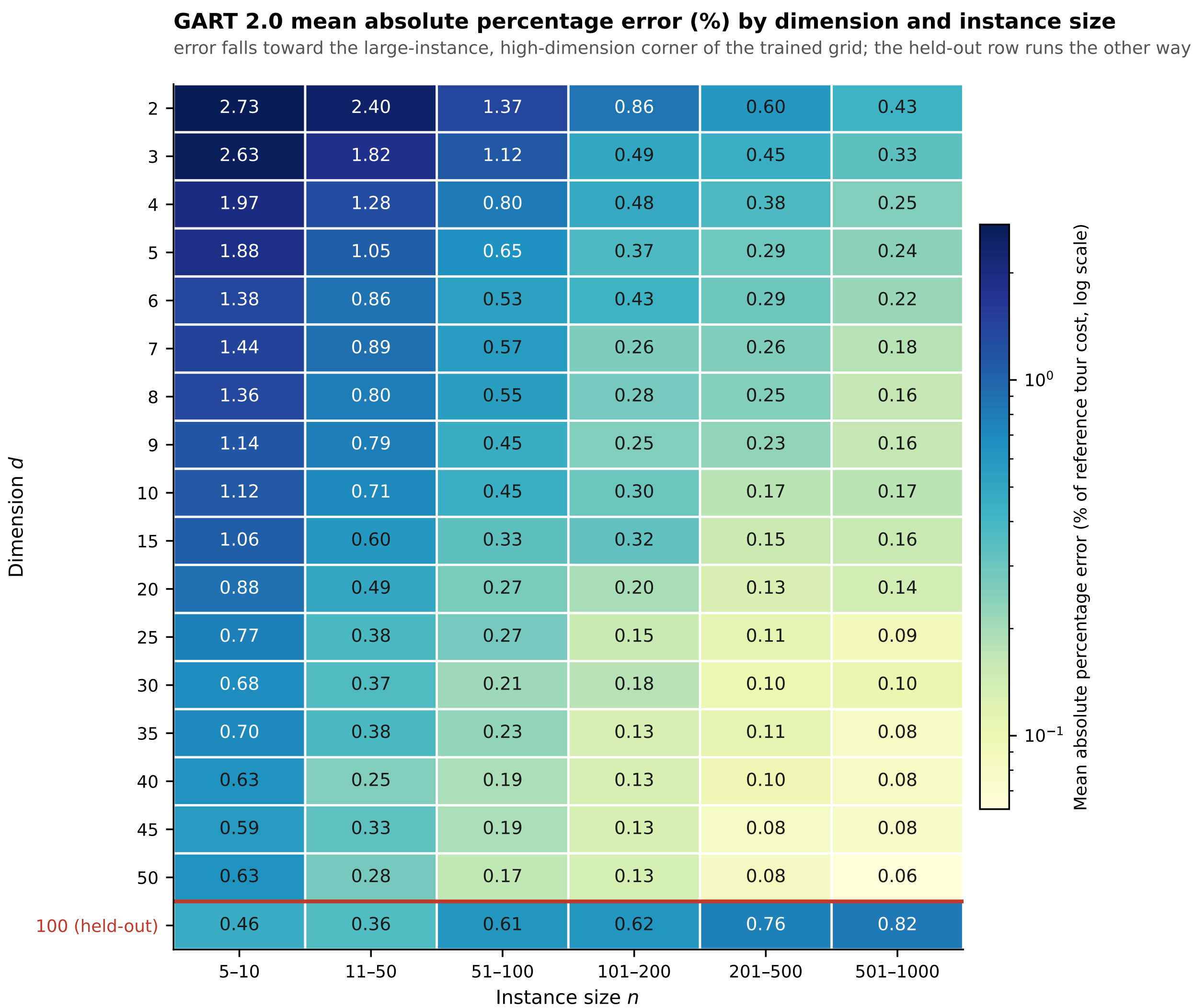

Figure 2: GART 2.0 MAPE by dimension and instance-size band; the red rule marks the held-out $d = 100$ row.

BHH's 28.01% error is not a defect of the reimplementation. The dominant term is the constant: for $d \geq 4$ the scored form substitutes a conjectured large-d asymptotic for β_d (Appendix F), and the row's signed error is negative at every

$d \geq 3$, the direction an understated constant produces. Two premises of the theorem fail here as well: only 68.7% of axes are uniform, so the region measure G^d exceeds the effective support of the rest — visible as the positive signed error at $d = 2$ — and $n \leq 1000$ with $d \geq 4$ is far from the asymptotic regime the theorem describes, where the sample must fill the region. Supplying the exact region rather than the convex hull nonetheless cuts its MAPE from 39.07% to 28.01%, which isolates how much of the convex-hull form's error is an input substitution; the convex-hull row itself is in the released tidy tables, not here, because at $d > 3$ that hull is replaced by the coordinate bounding box (Appendix F) and the row would mix two region definitions across dimensions.

4.3.2 2D Diverse Benchmark

Its 2,580 instances fall in five distribution classes, each probing a different geometric regime; Figure 3 shows the common-coverage error distributions.

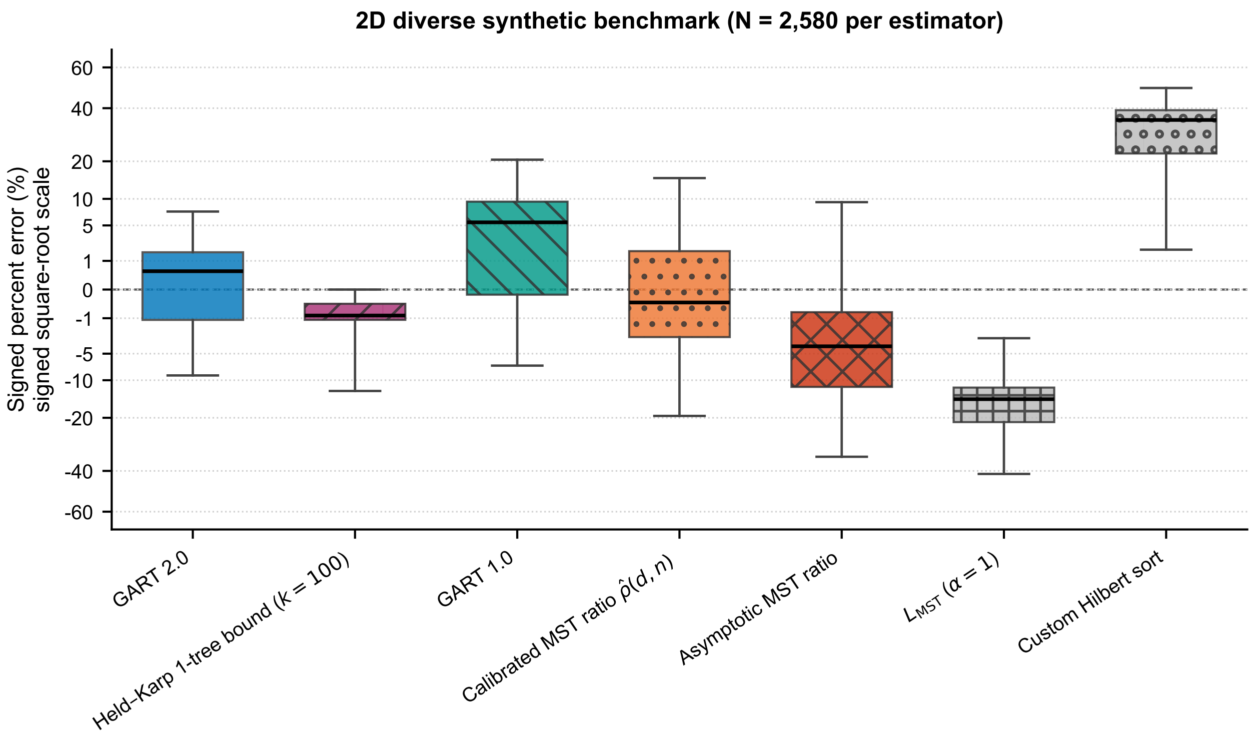

Figure 3: Signed percent error on the diverse 2D benchmark ($N = 2{,}580$ per estimator); axes and bound budget as in Figure 1.

GART 2.0 obtains 2.89% MAPE and 4.68% SDPE overall, against 5.76%/9.38% for the calibrated ratio $\hat{\rho}(d, n)$, 7.30%/7.90% for GART 1.0, 9.20%/11.55% for the asymptotic ratio, and 17.95%/10.27% for the $\alpha = 1$ floor. Its SDPE stays below 5.9% in every size bucket and falls to 2.75% in the $[501, 1000]$ bucket, and its aggregate is less than half the asymptotic ratio's. The custom Hilbert sort has 30.95% MAPE against 14.19% SDPE, a gap that identifies its error as a near-constant positive offset and not scatter, the signature a space-filling-curve tour should produce.

GART 2.0 ranges from 1.47% MAPE on the isotropic class to 10.82% on Line Noise, and every estimator degrades on Line Noise in the same direction: the calibrated ratio moves from 3.31% to 20.45%, the asymptotic ratio from 7.12% to 25.88%, and the $\alpha = 1$ floor from 17.15% to 34.13%. Near-collinear point sets drive α toward its upper bound of 2, and no estimator anchored on a typical ratio tracks that. Line Noise is not the only unrepresented geometry, and the other one fails in the opposite direction. On the 210 `grid` instances GART 2.0 reads 7.07% MAPE against 1.96% on the 420 Geometric Struct. instances that are represented, and its MSPE equals that MAPE exactly, so every one of those predictions is high. Realized α averages 1.05 on `grid` against 1.56 on Line Noise, so the two unrepresented generators sit at opposite ends of the target range and the estimator errs toward the middle on both. Both therefore confound unseen geometry with an extreme target value. On `grid` the $\alpha = 1$ floor is more accurate than the released model, at 4.54%, the only row in this paper on which the floor beats it.

4.3.3 TSPLIB95 EUC_2D Benchmark

This benchmark is the transfer test. Its instances come from real geography and circuit layouts and have no sampling region.

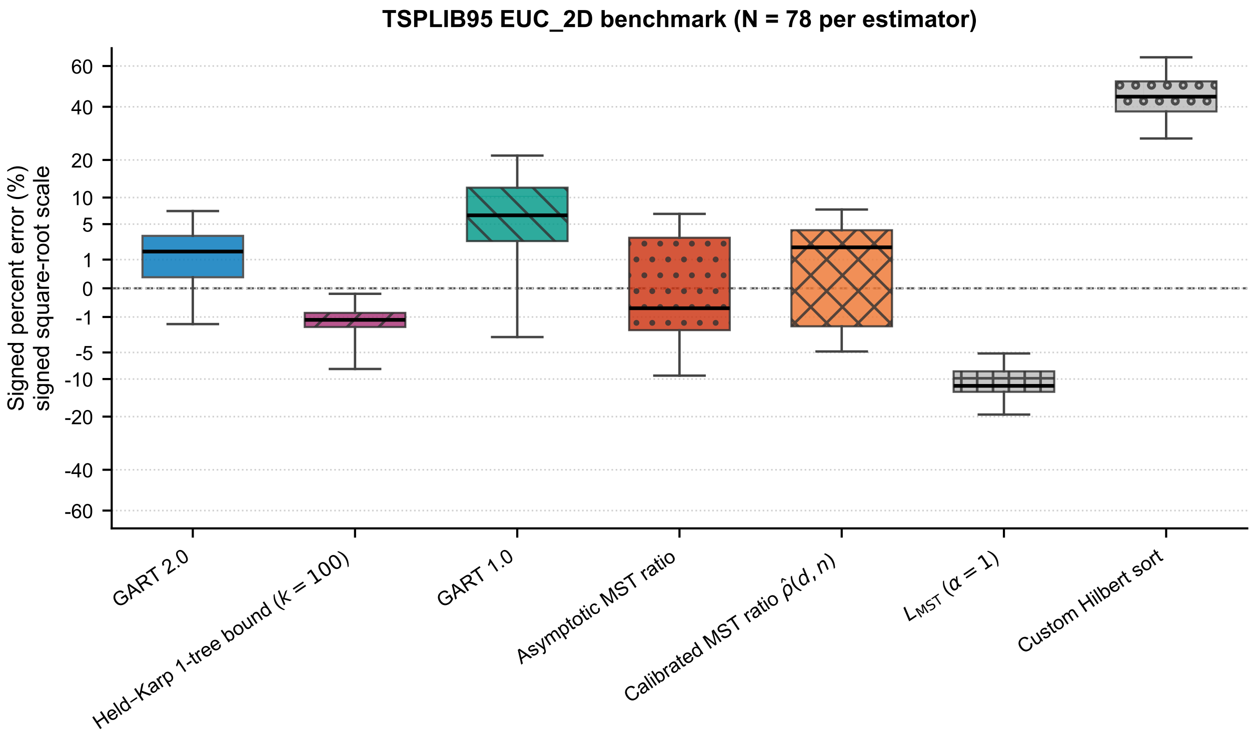

Figure 4: Signed percent error on the TSPLIB95 EUC_2D benchmark ($N = 78$ per estimator); axes and bound budget as in Figure 1.

On the full EUC_2D benchmark GART 2.0 obtains 2.53% MAPE and 2.92% SDPE, against 3.74%/4.48% for the calibrated ratio $\hat{\rho}(d, n)$, 3.53%/4.74% for the asymptotic ratio, and 8.43%/7.45% for GART 1.0. Table 17 reports all 78 instances, spanning $n = 51$ to 18,512, in three prespecified size buckets (23/16/39 instances), and Figure 4 shows the error distributions; the coarse $n > 400$ bucket should not be read as a homogeneous asymptotic regime. This is the narrowest margin in the paper, and it is narrow for a structural reason: α on these 78 instances has mean 1.1306 and standard deviation 0.0558, so a single well-chosen constant is already close. The MAPE-minimizing constant on these instances is 1.1275 and reaches 3.52%, so no constant multiplier does better on MAPE, and GART 2.0's 2.53% improves on that bound by 0.99 percentage points. The dispersion gap is the larger one, 2.92% SDPE against 4.74%, but SDPE alone cannot carry that comparison: for $\hat{L} = cL_{\text{MST}}$ the signed error is $c/\alpha - 1$, so SDPE is proportional to c and shrinking c drives it to zero at ruinous MAPE. The comparison holds because GART 2.0 attains the lower SDPE at the lower MAPE. GART 2.0's errors and the certified bound's are comparable in magnitude and opposite in direction. Its interquartile range lies wholly above zero and the bound's lies wholly below (Figure 4).

In the coarse $n > 400$ bucket the GART 2.0 and asymptotic-ratio SDPE intervals overlap, on $[3.20, 3.62]$. In the post hoc $n > 10{,}000$ slice of five instances GART 2.0 is lower on both metrics, 0.57% MAPE and 0.72% SDPE against the asymptotic ratio's 2.74% and 1.46%. The largest instance, `d18512`, is more than $18\times$ the training-size cap. For i.i.d. points of bounded density, normalized TSP and MST lengths each converge to dimension-dependent constants [Beardwood et al., 1959, Steele, 1988], which motivates a fixed-ratio baseline but does not establish that the heterogeneous $n > 400$ bucket is asymptotic; the advantage at extreme sizes is an observation on five instances and not proof of a convergence mechanism.

Both classical estimators are far behind here. Çavdar–Sokol obtains 23.87% MAPE and BHH 25.14%. TSPLIB instances carry no uniform density on a known region, so each area-based estimator runs outside the conditions it was derived under. BHH receives a convex hull in place of a sampling region that does not exist. Çavdar–Sokol is evaluated on graphs without a node at the corners of their covering rectangle, a property every one of its training graphs had. Both overpredict, by $+13.66\%$ and $+20.07\%$ respectively. These rows measure the cost of that mismatch, not the quality of the cited methods. Section 4.4 evaluates both where their assumptions hold.

4.4 Matched-Domain Comparison

Each classical estimator is scored here on the domain its published assumptions describe. The 210 instances of the `random` generator, the Uniform member of the Isotropic class of Table 2, are drawn i.i.d. uniformly on $[0, G]^2$, which is exactly the sampling model BHH assumes, and the region measure $A = G^2$ is known exactly. Çavdar–Sokol consumes no region, so it carries the same single row here that it carries above. Between the two panels only its instance set changes.

BHH falls from 23.80% MAPE on the full 2D benchmark to 8.91% here. The region input and the instance set both change between its two panels, so that figure combines the two effects and does not isolate the region input. What is left is a systematic underprediction, -8.65% signed against 7.76% SDPE, as an asymptotic constant evaluated at n as low as 5 should. Çavdar–Sokol falls from 18.24% to 8.16%, and here only the instance set changed, so that figure is a clean domain effect. It has the lower MAPE of the two classical estimators on every panel of Table 16, and the lower median absolute error here, but not the lower dispersion, and it is also the least even: at 14.28% SDPE its dispersion is the widest in the panel, and its median absolute error of 2.84% sits at just over a third of its mean, so it is close on most uniform instances and badly wrong on a few.

GART 2.0 beats both classical estimators on the matched domain, on both metrics, by a smaller and more credible margin than the full-benchmark tables suggest (Table 16). On the 210 uniform instances it obtains 1.31% MAPE against Çavdar–Sokol's 8.16%, BHH's 8.91% and the $\alpha = 1$ floor's 15.60%.

4.5 Generalization Beyond the Training Grid

The multidimensional test split is held out by row, not by geometry: it shares the four generators and the 68.7% uniform axis mix of the training split, and it is stratified on the same $(d, n,$ grid size$)$ cells. Three refits hold the hyperparameters and seed fixed and vary only the training data. A baseline refit reproduces the released model bit-for-bit at 1,118 trees, reading 0.6177% MAPE over the 16,846 scored rows.

Withholding both $d = 15$ and $d = 25$ from training, 11.8% of the training rows, raises test MAPE at $d = 15$ from 0.50% to 0.58% and at $d = 25$ from 0.35% to 0.38%, while the neighboring trained dimensions $d \in \{10, 20, 30\}$ do not degrade at all, moving from 0.43% to 0.42%. The loss is specific to the withheld strata rather than to the reduced training volume, and it takes the form of a bias shift: mean signed error at $d = 15$ moves from -0.008% to $+0.134\%$. Training only on $n \leq 200$ and testing on $n \in (200, 1000]$ raises MAPE from 0.41% to 0.59%, again almost entirely as a uniform offset, since SDPE moves only from 0.47% to 0.56% while mean signed error rises from $+0.28\%$ to $+0.51\%$, a factor of 1.8.

Neither held-out regime degrades by more than two tenths of a MAPE point against its matched in-grid baseline. The estimator does exploit having seen a cell of the grid. Off the grid it acquires a systematic offset and keeps its structure. We demonstrate no correction for that offset.

4.6 Failure on Near-Collinear Geometry

Section 4.3.2 reports 10.82% MAPE on the near-collinear Line Noise class against 1.47% on the isotropic class, and the error is almost entirely one-sided: MSPE is -10.82, so the model under-predicts. Two explanations compete. The training split may never show the model this regime, or the feature set may be unable to represent it. Both hold.

The support gap is exact. The training split contains zero instances with $\alpha > 1.5$ at $n \geq 200$, and every training instance with $\alpha > 1.7$ has $n \leq 10$. Its α distribution has a 99th percentile of 1.544, and 54% of the 210 benchmark Line Noise instances lie above that percentile. No tree can map large n at high α onto a high $\hat{\alpha}$ when the fit never contains the combination. The symptom matches: regressing predicted α on true α over the 90 Line Noise instances at $n \geq 200$ gives a slope of 0.36 against a correlation of 0.95, so the model ranks those instances almost perfectly and declines to move its α level.

A 2×2 ablation separates the two causes. Holding hyperparameters and seed fixed, we vary the feature set, extending the 30-feature baseline with five degeneracy, local-intrinsic-dimension and MST-topology descriptors, and we vary the training support, adding 874 newly generated and solved instances at large n and high α to the training split only. Measured by the Line Noise slope at $n \geq 200$, features alone move the baseline from 0.294 to 0.569, support alone moves it to 0.332, and the two together reach 0.798. Neither intervention approaches sufficiency alone and the two are not additive, so the deficit is joint. The ladder is anchored on the 30-feature model, whose slope on this slice is 0.294; the released 31-feature model reaches 0.362 on the same slice, so these figures compare the two interventions against each other rather than against the released estimator.

We attempted the repair twice against criteria fixed before either candidate was scored, and rejected both. Adding all 874 instances to the training split clears every one of nine pre-registered gates, moving 2D MAPE from 2.89% to 2.04%, Line Noise MAPE from 10.82% to 5.83% and the slope from 0.366, that experiment's own baseline reading, to 0.838, but two defects disqualify it. The slope is not a property of the model: eight permutations of the row order of the identical training rows, at one seed, span $[0.637, 0.803]$, and the reported 0.838 lies above that entire band. And the gain is not a cross-family result, because the augmentation's $d = 2$ lattice generator and the benchmark's `grid` generator place their sites, spacing, cell centers and $\pm 5\%$ jitter identically, so half of the 2D gain is measured on the candidate's own training geometry. A de-contaminated second candidate, re-run over seven seeds fixed in advance, removes that overlap and clears ten of its own eleven gates at a median slope of 0.735, but forfeits the paper's one distribution-independent dispersion claim to do so: its Holm-adjusted advantage over the asymptotic ratio on TSPLIB reaches only $p = 0.065$ at the median against 0.0048 for the released model. We did not make that trade. A per-family multiplicative constant, an oracle a deployed estimator does not have, recovers 97.9% of the first candidate's Line Noise gain by itself, so most of what either candidate buys is a per-family offset rather than a repaired response to geometry. The degenerate-geometry failure is therefore diagnosed and not repaired, and the released model is unchanged.

4.7 Rank Agreement and Calibration

Many of the downstream uses in Section 1 instead compare instances and act on the comparison, asking which candidate node set is cheaper or which route split to keep. The estimator must then *order* instances the way the stored references do even when its absolute numbers are off. The close-pair statistic below is a plain concordant fraction and sits at 0.5 under an unrelated ordering. Global rank correlation is close to 1 for every estimator here, so it separates almost nothing. Cross-instance cost varies over orders of magnitude and L_{MST} already tracks that variation, so a high rank correlation demonstrates nothing about the learned ratio. Both metrics are reported with L_{MST} itself as the control. Every constant multiplier is a monotone transform of L_{MST} and scores identically to it, so the asymptotic-ratio and $\alpha = 1$ controls coincide on 2D and TSPLIB.

Close-pair ordering is the discriminating measure. For pairs satisfying $|L_A - L_B| / \max(L_A, L_B) < 5\%$, GART 2.0 orders 74.5% of 74,826 2D pairs, 92.4% of 1,647,719 multidimensional pairs, and 72.2% of 54 TSPLIB pairs correctly, against 71.0%, 69.1%, and 61.1% for the $\alpha = 1$ control. The multidimensional gain of 23.4 percentage points is the substantive one; the 2D gain of 3.4 points is not, and on TSPLIB 54 pairs cannot support an 11-point claim. At a 10% threshold GART 2.0 obtains 83.6%, 96.1%, and 81.0%. The percentages remain descriptive because the pairs share instances and no block-bootstrap interval is computed over them. GART 1.0 trails GART 2.0 on the TSPLIB close-pair statistic at the tighter threshold, at 70.4%. Cost accuracy and pair ordering are distinct properties, and 54 pairs resolve none of these gaps.

Two calibration caveats bound the cost-level results. The $n > 400$ TSPLIB bucket has $R^2_\alpha = -0.071$: the model predicts α worse than the bucket mean does. That bucket has low target variance and an upward mean prediction offset while its tour-cost MAPE remains 2.90%, so the cost-level result does not remove the ratio-calibration failure. The aggregate margins are also not uniformly large. We test each comparison on the instances where both estimators produce a finite prediction, using the paired difference in absolute percent error with a 1,000-resample paired bootstrap and a Wilcoxon signed-rank test. On TSPLIB EUC_2D the difference between GART 2.0 and the asymptotic MST ratio is -1.00 percentage points with a 95% interval of $[-1.77, -0.22]$ and $p = 0.0049$: on that benchmark GART 2.0's MAPE advantage is small but resolvable, and its larger gain is dispersion, 2.92% SDPE against 4.74%.

4.8 Computational Cost

Feature extraction dominates the estimator's cost. In the TSPLIB timing log GART 2.0 spends 92.90% of its wall time on feature extraction and MST construction against 5.03% on LightGBM inference, so the tree traversal is not the target for any speed work and the ensemble size is not what the cost claim turns on. The Time column of Table 17 puts every estimator on one protocol: one estimator per process, a single thread, the median of 11 repeats. On that protocol GART 2.0 costs 6.12 ms over the full EUC_2D set, matching GART 1.0, roughly twice the 2.57–3.08 ms of the rows that build an MST or a Hilbert sort, and roughly an order of magnitude over the 0.645–0.880 ms of the two classical closed forms, which compute a convex hull and then evaluate an algebraic expression. That ordering is the price of the feature set and it is stable across the size range; at $n > 400$ GART 2.0 is the most expensive row in the table at 20.54 ms.

Those multiples are the wrong comparison for deployment, where the alternative is a solve. On the multidimensional benchmark the median prediction and the median reference-tour generation cost the same, 121 ms against 122 ms, since most of those instances are small enough to solve outright and an estimator buys nothing. The argument holds only where the solve is expensive, and there it holds decisively: at $n \in [501, 1000]$ the reference tour takes 3,799 ms against GART 2.0's 460 ms, and on TSPLIB `d18512`, at $n = 18{,}512$ and more than $18\times$ the largest training instance, GART 2.0 predicts in 239 ms.

5 Comparison Against a Certified Lower Bound

Every baseline of Section 4.1 is an estimator. It returns a number with no guarantee, scored only by how close it lands. The Held–Karp 1-tree bound is not. On the aggregate of all four benchmarks it is cheaper and more accurate than GART 2.0, on TSPLIB EUC_2D once scaled by a single training-split constant, so the estimator is off the cost/accuracy Pareto front of all four. However, the aggregates conceal how each family scales in n. The ascent is quadratic and the planar estimator is not.

5.1 The Held–Karp 1-tree Bound

Fix a node s. A *1-tree* is a spanning tree on $V \setminus \{s\}$ together with any two edges incident to s; it has n edges and total degree $2n$, and every Hamiltonian tour is a 1-tree, so the minimum 1-tree — a minimum spanning tree on $V \setminus \{s\}$ plus the two cheapest edges at s — is a relaxation of the tour problem. Introduce node potentials $\pi \in \mathbb{R}^n$ and transformed costs $c^\pi_{ij} = c_{ij} + \pi_i + \pi_j$. Under c^π every tour's cost shifts by exactly $2\sum_i \pi_i$, because every node has degree two, so the optimal tour under c^π costs $L_{\mathrm{TSP}} + 2\sum_i \pi_i$ and for *any* real π

$$w(\pi) \;=\; L_{\mathrm{1tree}}(\pi) \;-\; 2\sum_{i\in V}\pi_i \;\le\; L_{\mathrm{TSP}}, \tag{2}$$

which is the Held–Karp bound [Held and Karp, 1970, 1971]. The map w is concave and piecewise linear, a subgradient at π is $g_i = \deg_i(\pi) - 2$, and $\max_\pi w(\pi)$ is approached by subgradient ascent. Subgradient ascent is not monotone in w, so the scored quantity at budget k is the incumbent $\hat{w}_k = \max_{j\le k} w(\pi_j)$. The budget k trades cost against tightness, and on any one ascent the incumbent is non-decreasing in k by construction. Validity needs neither $\pi \ge 0$ nor $c^\pi \ge 0$: the argument above is sign-free, and the spanning tree is built on the perturbed costs directly.

Eq. (2) holds at every π, so $\hat{w}_k$ is a *certificate* whatever the ascent did. The printed number is a proven lower bound on that instance's optimal tour, and a reader can verify it from the 1-tree alone. No other row in this paper certifies its own output, GART 2.0 included. The bound also consumes no training corpus, so no question of coverage, distribution shift, or leakage arises.

The certificate is one-sided. Because $\hat{w}_k$ never exceeds the optimum, its error against a reference tour is a duality gap plus whatever that reference sits above the optimum. The *calibrated* bound $c_k\,\hat{w}_k$ is also reported, where c_k is a single scalar fitted per budget on the planar training split and never on any scored instance. That row is not a certificate, since $c_k > 1$ pushes some predictions above the optimum, and it is not training-free. It is the like-for-like comparator against a trained estimator, and it is the stronger of the two rows on every corpus that carries it. Its constant is planar-fitted, so the multidimensional ladder reports the raw bound alone.

Both ascents run on all four benchmarks, and each headline table reports the arm that leads on its own corpus. The planar tables use the Volgenant–Jonker step schedule with Helsgaun's direction smoothing [Volgenant and Jonker, 1982, Helsgaun, 2000]. The multidimensional tables use a Polyak step [Polyak, 1969] against a constructive upper bound, a nearest-neighbor tour improved by 2-opt and built from coordinates alone. Neither ascent is proved to attain $\max_\pi w(\pi)$, so every accuracy figure below is a floor on what this family reaches.

5.2 Results on TSPLIB95 EUC_2D

Cost is measured on the protocol of Table 17, one estimator per process inside a single quiet window, and both methods are compared on the 77 of the 78 EUC_2D instances the bound's dense kernel covers; the one excluded instance is reported at the end of this subsection. The ms column is the median per-instance time, and the accuracy beside it is a MAPE and therefore a mean. $\times$ compares the bound's time with GART 2.0's. It is the ratio of the printed medians here and on the 2D and non-Euclidean ladders, and the median of back-to-back per-instance ratios on the multidimensional ladder. Raw is the certified incumbent $\hat{w}_k$. Cal. is $c_k\hat{w}_k$ with c_k fitted per budget on the planar training split. Bold marks every cell both cheaper and more accurate than GART 2.0. A *rung* is one budget row.

Read on the corpus median, the raw certified bound first matches GART 2.0's MAPE at an ascent budget of 50, where it costs 1.46 times GART 2.0 for a margin of 0.0041 percentage points, and where the paired win rate is exactly 50.0%. At a budget of 25 the calibrated row costs 0.90 times GART 2.0 and reaches 2.00% MAPE against 2.56%: strict domination on both axes, on the same instances and the same protocol. That multiple is a ratio of typical-instance costs; summed over the corpus instead, the same pair reads 24.94, because the bound is cheap on the small instances that set the median and dear on the large ones that set the total. The ladders report the first statistic. GART 2.0 is therefore not on the cost/accuracy Pareto front of this corpus median once the family is admitted.

On the 23 smallest instances the calibrated bound at that same budget costs 0.16 times GART 2.0 at 1.69% MAPE against 2.01%, and even the raw uncalibrated certificate reaches 22.2% lower error at 0.46 times the cost. In the middle bucket nothing dominates, and the raw bound needs the top of the ladder to match at all. In the largest bucket the raw bound matches at 6.65 times the cost and again nothing dominates. The reversal is monotone in n: the comparison yields a rule keyed to instance size.

The bound is far more load-sensitive than GART 2.0, its median rising by a factor of 1.25 between a quiet and a noisy window at the crossing budget of 50 against 1.05 for GART 2.0, so every published repeat is from one quiet window and the noisy repeats are retained only as the control; mixing the two would have overstated the bound's cost and flattered GART 2.0. And the excluded instance is real cost, not a hidden exclusion: on `d18512`, above the bound's dense kernel switch, the bound costs 167 times GART 2.0 at the crossing budget and 2,078 times at the top of the ladder, 493 seconds against 237 milliseconds (the 239 ms of Section 4.8 is the same work under the harness protocol).

Table 3: TSPLIB95 EUC_2D cost/accuracy ladder over the 77 matched instances; column conventions are defined at the start of this subsection.

	$n \in [51, 150]$, $N = 23$				$n \in [151, 400]$, $N = 16$				$n > 400$, $N = 38$			
Row	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.
GART 2.0	3.73	1.00	2.01		4.55	1.00	2.38		18.29	1.00	2.97	
$k = 0$	0.19	0.05	11.83	3.12	0.45	0.10	11.31	3.59	24.49	1.34	9.13	4.01
$k = 10$	0.37	0.10	8.47	2.04	0.91	0.20	9.85	3.76	43.93	2.40	7.63	2.53
$k = 25$	0.59	0.16	2.91	**1.69**	1.59	0.35	4.84	3.09	72.52	3.97	3.48	1.72
$k = 50$	0.96	0.26	2.09	**1.68**	2.79	0.61	3.82	3.03	121.59	6.65	2.31	1.56
$k = 100$	1.72	0.46	**1.56**	**1.20**	5.16	1.13	2.97	2.29	222.22	12.15	1.96	1.33
$k = 200$	3.22	0.86	**1.57**	**1.22**	10.10	2.22	2.96	2.28	427.85	23.40	1.88	1.26
$k = 500$	7.88	2.11	0.97	0.74	25.65	5.64	1.93	1.42	1120.37	61.27	1.62	1.12

5.3 Results on the Multidimensional Benchmark

Table 4 reports the ladder by dimension group. At an ascent budget of 200 the bound reaches 0.111% MAPE against GART 2.0's 0.618%, a factor of 5.56, at 0.71 times the cost, and it wins the paired comparison on 82.9% of instances. It first overtakes GART 2.0 at a budget of 100, where it costs 0.40 times as much. Reweighting the dimension-stratified timing sample to the corpus composition puts the cost ratio at a budget of 200 at 0.54.

The domination holds at $d \in [4, 10]$, at $d \in [15, 50]$ and at $d = 100$, and it is widest exactly where the estimator was never trained: at $d = 100$ a budget of 500 costs 0.35 times GART 2.0 and is 140.8 times more accurate. The one group GART 2.0 holds is $d \in \{2, 3\}$, where matching its accuracy costs the bound 1.27 times as much.

It is, however, an artifact of instance size, and the aggregate above conceals that entirely. Of the 16,846 scored instances 10,569, or 62.74%, carry at most 100 nodes, a regime in which the relaxation is very nearly exact and very nearly free; a corpus mean over that composition is largely a statement about small instances rather than about the estimator. We draw 20 instances per (dimension, size band) cell, matched on size across dimensions so that the cost ratio reports a dimension effect and not a size effect, and price the ladder on them under the protocol of Table 3; accuracy in each cell is measured on every instance both methods score, between 108 and 2,209 of them. GART 2.0 is undominated in 10 of the 18 cells, and the eight in which the bound is both cheaper and more accurate are the six at $n \in [20, 100]$ together with the two at $d = 100$. At $n \geq 200$ the estimator is on the cost/accuracy front at every dimension it was trained on, and it is dominated only at $d = 100$, the dimension withheld from training, where its own accuracy degrades to 0.824% against 0.062% at $d = 50$.

In the $n \in [600, 1000]$ band GART 2.0 reaches 0.429% MAPE at 10.0 ms at $d = 2$, where the tightest rung we priced, $k = 500$, is both dearer and looser at 0.784% and 246.1 ms; 0.332% at 25.7 ms at $d = 3$, against 0.450% at 143.9 ms for $k = 200$; and 0.251% at 58.2 ms at $d = 4$, against 0.291% at 86.2 ms for the same budget. The mechanism is the one Section 5.5 states: the bound is $\Theta(kn^2)$ in every dimension, so the cost of matching a fixed accuracy grows quadratically while the estimator's cost carries no accuracy term, and outright reversal — GART 2.0 dominating — arrives only in the plane and at $d = 3$, from roughly 200 nodes upward.

The ascent never reads the stored label. Supplying the released optimum in place of the coordinates-only upper bound converges faster early and to the same value. On the 3,069 scored instances whose label the generation run recorded as a

Concorde-proven optimum — a record of that run, not a certificate the released artifacts retain (Section 3.2) — the target is an optimum rather than a heuristic tour, and GART 2.0 scores 1.03% against the bound's 0.046% at a budget of 200. And the relaxation closes exactly, returning the optimum rather than a bound, on 37.2% of the scored multidimensional split. Pairwise distances concentrate as d grows, so the integrality gap this relaxation is usually described by is a planar fact.

Roughly one instance in seven exhausts the largest budget we ran, so at the top of the size range the reported accuracy is a floor. The cost figure in each cell is a median over 20 timed instances against an accuracy figure over the whole cell, so the cost column is the weaker of the two and a cell whose verdict turns on a few milliseconds should be read as a trade-off rather than as a ranking; budgets that close their gap terminate early, which is why a larger budget can occasionally read cheaper by a fraction of a millisecond. And the estimator's own asymptotics change with dimension: `compute_mst` dispatches to the Delaunay construction only for $d \leq 3$ and to a dense kernel from $d = 4$ upward, so above three dimensions both families are quadratic in n and the separation is a constant rather than an exponent. That constant favors the estimator at $n \geq 200$ and the bound below it, the quantity Table 5 measures; it is not a complexity argument and we do not extrapolate it past $n = 1000$, the largest size this corpus carries.

Table 4: Multidimensional cost/accuracy ladder, all 16,846 scored instances, Polyak ascent; conventions as in Section 5.2. Only the raw bound is reported, the calibration constant being planar-fitted.

	$d \in \{2,3\}$, $N = 1{,}295$			$d \in [4,10]$, $N = 4{,}534$			$d \in [15,50]$, $N = 5{,}170$			$d = 100$, $N = 5{,}847$		
Row	ms	×	MAPE	ms	×	MAPE	ms	×	MAPE	ms	×	MAPE
GART 2.0	5.87	1.00	1.453	3.30	1.00	0.728	3.59	1.00	0.340	3.26	1.00	0.593
$k = 0$	2.43	0.24	12.190	0.13	0.04	7.446	0.37	0.07	3.834	0.23	0.07	2.133
$k = 10$	3.51	0.31	7.389	0.24	0.08	4.500	0.36	0.12	2.164	0.26	0.08	1.026
$k = 25$	5.18	0.47	4.746	0.38	0.13	3.240	0.55	0.17	1.461	0.37	0.12	0.691
$k = 50$	7.37	0.69	3.177	0.62	0.21	1.972	0.92	0.27	0.870	0.49	0.15	**0.412**
$k = 100$	12.90	1.27	1.388	1.21	0.39	**0.692**	1.95	0.56	**0.301**	0.76	0.23	**0.147**
$k = 200$	22.97	2.18	0.557	2.15	0.69	**0.161**	3.30	0.95	**0.053**	1.19	0.37	**0.025**
$k = 500$	50.80	4.92	0.384	4.51	1.38	0.067	6.05	1.75	0.011	1.12	0.35	**0.004**

5.4 Results on the 2D Diverse and Non-Euclidean Benchmarks

The remaining two ladders are measured on the protocol of Section 5.2, with one timing exception disclosed below.

Table 6 gives the 2D diverse ladder. The raw certified bound passes GART 2.0 under the Volgenant–Jonker step at a budget of 50, at 0.23 times the cost with a paired win rate of 54.8%; the calibrated row passes it a rung earlier. The reversal is monotone in n and steeper than the one Section 5.2 reports, because this corpus reaches down to five nodes. On the 360 instances with $n \leq 10$, the raw bound at a budget of 25 reads 0.439% MAPE against GART 2.0's 3.212% at 0.07 times the cost, and wins 95.0% of the paired comparisons. On the 610 instances with $n > 500$, nothing on the ladder dominates: the raw bound is more accurate than GART 2.0 only at the top budget, and there it costs 26.08 times as much.

Table 7 gives the non-Euclidean ladder, where the estimator needs the MDS embedding of Section 6 and the bound reads the distance matrix directly. Over the 29 instances both methods score, GART 2.0 costs 4.88 ms, of which the embedding is 9.21%, at 3.814% MAPE. The Polyak step at a budget of 500 reaches 0.509% MAPE at 0.69 times that cost and wins all 29 paired comparisons; at a budget of 200 it reads 0.762% for 0.34 times the cost, again on every instance. The cheapest domination by a certified row anywhere in this study is on this corpus: the Volgenant–Jonker step at a budget of 25 costs 0.065 times GART 2.0 and reads 2.953% against 3.814%.

Table 5: The multidimensional frontier stratified by size and dimension; "Dominates" means strictly better on cost and accuracy at the listed budgets. The bound is the Polyak arm.

d	n band	N	GART 2.0 MAPE (%)	GART 2.0 (ms)	Verdict against the ladder
2	[20, 100]	243	1.829	2.9	bound dominates ($k = 200,\ 500$)
	[200, 500]	108	0.663	5.7	**GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200,\ 500$)
	[600, 1,000]	135	0.429	10.0	**GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200,\ 500$)
3	[20, 100]	243	1.431	3.4	bound dominates ($k = 100,\ 200,\ 500$)
	[200, 500]	108	0.459	12.2	**GART 2.0 dominates** ($k = 100,\ 200$)
	[600, 1,000]	135	0.332	25.7	**GART 2.0 dominates** ($k = 25,\ 50,\ 100,\ 200$)
4	[20, 100]	243	1.015	2.4	bound dominates ($k = 200$)
	[200, 500]	108	0.403	14.2	trade-off at every rung
	[600, 1,000]	135	0.251	58.2	**GART 2.0 dominates** ($k = 200$)
10	[20, 100]	243	0.566	2.4	bound dominates ($k = 200$)
	[200, 500]	108	0.204	14.7	trade-off at every rung
	[600, 1,000]	135	0.173	100.2	trade-off at every rung
50	[20, 100]	243	0.214	2.8	bound dominates ($k = 200$)
	[200, 500]	108	0.095	22.7	trade-off at every rung
	[600, 1,000]	134	0.062	114.3	trade-off at every rung
100	[20, 100]	2,209	0.501	3.4	bound dominates ($k = 100,\ 200$)
	[200, 500]	977	0.723	26.8	bound dominates ($k = 50,\ 100$)
	[600, 1,000]	1,185	0.824	120.4	bound dominates ($k = 50,\ 100$)

The two step rules do not order the same way on the two corpora. Volgenant–Jonker leads Polyak at every 2D budget that runs an ascent except 200, where the two are separated by half a hundredth of a percentage point; the gap is widest at a budget of 25, 3.606% against 7.515%. Polyak leads from a budget of 200 on the non-Euclidean corpus, 0.762% against 1.228%.

The timing sessions behind both ladders did not run on a quiet machine, and the two methods are not equally sensitive to that: a paired control against the published quiet-window measurements puts GART 2.0 at 1.48 times its published cost and the bound at 1.10 to 1.14 times, so every cost multiple above is between 0.74 and 0.77 of its quiet-box value. That direction runs against the bound, and correcting for it costs it exactly one domination, the raw Volgenant–Jonker row at a budget of 200 on the 2D corpus, whose 0.79 becomes 1.03. And the checkpointed protocol does not flatter the bound: running the released single-budget call once per rung, sharing nothing between rungs, costs 0.77 to 0.95 times the checkpointed ladder on the 2D corpus and 0.90 to 1.14 times on the non-Euclidean one.

Table 6: 2D diverse cost/accuracy ladder, all 2,580 instances, both step rules.

	Volgenant–Jonker step				Polyak step			
Row	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.
GART 2.0	5.00 ms at $1.00\times$; 2.889% MAPE							
$k = 0$	0.23	0.046	13.938	5.654	0.31	0.062	13.938	5.654
$k = 10$	0.44	0.087	8.352	4.257	0.49	0.098	10.572	6.872
$k = 25$	0.70	0.14	3.606	**2.564**	0.73	0.15	7.515	6.393
$k = 50$	1.15	0.23	**2.702**	**2.330**	1.12	0.22	5.526	4.978
$k = 100$	2.05	0.41	**2.048**	**1.763**	1.88	0.38	3.194	**2.788**
$k = 200$	3.95	0.79	**1.883**	**1.618**	3.36	0.67	**1.878**	**1.650**
$k = 500$	9.42	1.88	1.274	1.102	7.15	1.43	1.447	1.331

Table 7: Non-EUC_2D TSPLIB95 ladder over the 29 instances both methods score, nine of which violate the triangle inequality; conventions as in Section 5.2, except that the planar c_k makes the Cal. columns indicative only.

	Volgenant–Jonker step				Polyak step			
Row	ms	$\times$	Raw	Cal.	ms	$\times$	Raw	Cal.
GART 2.0	4.88 ms at $1.00\times$; 3.814% MAPE							
$k = 0$	0.03	0.007	15.606	6.181	0.05	0.011	15.606	6.181
$k = 10$	0.17	0.035	8.150	**3.189**	0.16	0.033	10.087	5.481
$k = 25$	0.32	0.065	**2.953**	**1.784**	0.28	0.057	6.501	5.053
$k = 50$	0.56	0.11	**1.783**	**1.436**	0.49	0.100	3.817	**3.129**
$k = 100$	1.10	0.23	**1.269**	**1.037**	0.92	0.19	**1.990**	**1.589**
$k = 200$	2.15	0.44	**1.228**	**1.018**	1.66	0.34	**0.762**	**0.581**
$k = 500$	5.27	1.08	1.139	0.995	3.38	0.69	**0.509**	**0.492**

5.5 Complexity of the Estimator Families

The classical closed forms are dominated by a row-deduplication pass rather than by their arithmetic. The convex-hull blow-up this paper cites as a motivation never occurs at runtime, because the implementation caps hull construction at three dimensions and falls back to a bounding box above it. And the 1-tree cannot use the Delaunay shortcut the MST family relies on in the plane. The node potentials break the Euclidean metric, so after the first iteration the perturbed minimum spanning tree is not a subgraph of the triangulation. At 144 potential vectors drawn from real ascent trajectories the Delaunay-restricted 1-tree is heavier than the exact one at 141 of them, which is why the bound sits a complexity class above the MST family and not a constant factor above it.

Figure 5 measures this class difference. In the plane GART 2.0 is $\Theta(n \log n)$ and the bound is $\Theta(n^2 d + kn^2)$, and the measured log–log cost exponents in n agree: 0.975 for GART 2.0 above a thousand nodes against 2.08–2.13 for the bound over the same window. GART 2.0 costs a flat 2.0 to 2.3 times the MST-ratio family across four orders of magnitude in n, which is the signature of a shared complexity class; against the 1-tree at a fixed budget the ratio moves from 7.7 at a thousand nodes to 211 at sixteen thousand. The bound's advantage at small n comes from constants. The estimator's advantage at large n comes from growth, and only the second is stable.

The separation is planar, and it does not extend further. On the multidimensional benchmark GART 2.0's own measured cost exponent in n is 2.02 at $d \in [4, 10]$ and 1.82 at $d \in [15, 50]$, against 1.785–1.889 for the bound, because the MST construction falls back to a dense all-pairs kernel from $d = 4$ upward. Only at $d \in \{2, 3\}$ is that exponent 0.56. On that benchmark both families are quadratic, the separation is a constant, and the constant points the wrong way.

Figure 5 adds an exact solve to the same axis, its curve leaving the chart where the wall-clock cap censors it. Branch-and-cut admits no polynomial bound, and the Held–Karp dynamic program [Held and Karp, 1962] is $\Theta(n^2 2^n)$, so every row of Table 8 is asymptotically cheaper than an exact solve and being cheaper than one distinguishes nothing among them. The comparison that discriminates is the 1-tree, which is polynomial, certified, and in the same cost decade as the estimators.

5.6 Operating Regime

GART 2.0 is cheaper than a 1-tree bound above roughly 400 nodes in the plane and dearer below roughly 150. Its accuracy sits between the bound's and the constant-ratio family's: against the MST-ratio family it is separated by 1.00 percentage points on the full 78-instance EUC_2D set, against 1.35 points from the converged bound on the same set.

On the aggregate of every benchmark here the bound is the better instrument. It does not follow that the estimator is superseded, because the multidimensional aggregate is a corpus 62.74% composed of instances of at most 100 nodes, and that is the regime the aggregate describes. Read at fixed size instead (Table 5), the ordering turns over: at $n \geq 200$

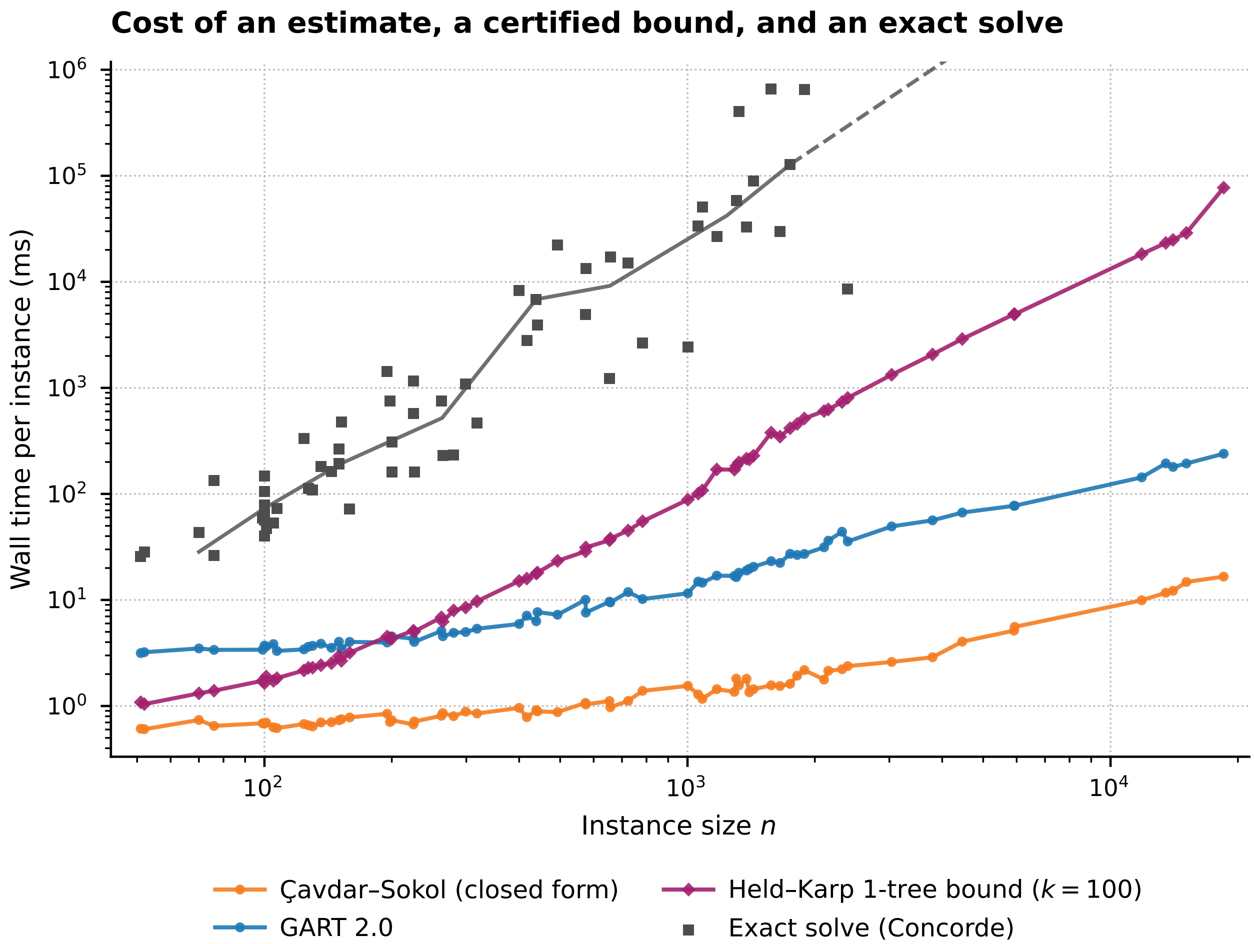

Figure 5: Per-instance cost against instance size on TSPLIB95 EUC_2D.

Table 8: Dominant per-instance cost of each family as the released code executes it.

Family	**Dominant term as implemented**	**Guarantee**	**Training**
Classical closed forms	$\Theta(d\,n\log n)$; a row-deduplication pass, not the arithmetic	None	Constants fitted in the source
Constant multiples of L_{MST}	$\Theta(n\log n)$ at $d=2$ (Delaunay); $\Theta(n^2 d + n^2\log n)$ otherwise	Lower bound only at $\alpha = 1$	The two calibrated rows only
Space-filling curve (Hilbert)	$\Theta(ndh)$ interpreted, plus an $O(n\log n)$ sort	Upper bound: it builds a tour	None
GART 2.0 (learned head on this feature block)	Feature cost as the MST family above, except the greedy tour's dense $\Theta(n^2)$ scan below its candidate-list threshold; head $O(1)$ in n	None	Yes
GART 1.0	$\Theta(n^2 d)$ time and $\Theta(n^2)$ memory; a dense matrix for two scalars	None	Yes
Held–Karp 1-tree, budget k	$\Theta(n^2 d + kn^2)$; no Delaunay path exists after the first iteration	Lower bound	None
Calibrated 1-tree	As above	None	One scalar per budget
Exact solver	$\Theta(n^2 2^n)$ for the Held–Karp dynamic program; branch-and-cut admits no polynomial bound	Exact	None

GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the two larger size bands it is strictly better than the bound on both axes from an ascent budget of 25 upward in the plane and over the mid-ladder budgets at $d = 3$.

Below roughly 200 nodes, prefer the certified bound at any dimension: it is cheaper, it is more accurate, it needs no training corpus, and it tells the caller how wrong it can be. Above that size prefer the estimator at any dimension it was trained on, most decisively in the plane and at $d = 3$, where the ordering not only reverses but keeps reversing, because each ascent iteration costs the bound $\Theta(n^2)$ at a fixed accuracy in every dimension while the estimator's planar cost stays subquadratic. At $d = 100$, outside the training grid, prefer the bound at every size; that is the one cell in Table 5 where large n does not rescue the estimator, and the reason is its own degradation off the grid rather than the bound's strength. Given only a distance matrix, prefer the bound at any size. It reads the matrix directly, and the estimator pays an embedding to reach it at all (Section 6). Where inputs are known to be i.i.d. uniform on a known region and an error in the high single digits of percent is acceptable, the classical closed forms remain the cheapest instrument in this paper by an order of magnitude, with no MST to build. Large instances re-evaluated many times under changing input are the last-mile and partitioning setting of Section 1. In that regime an exact solve is out of reach, a quadratic bound is already expensive, and GART 2.0's accuracy is at its best.

6 Application: Non-Euclidean Estimation via MDS Embedding

The preceding benchmarks use Euclidean point clouds, and the certified bound of Section 5 needs no more than a symmetric distance matrix. TSPLIB95 contains 33 non-EUC_2D instances: 4 CEIL_2D, 2 ATT, 10 GEO, and 17 EXPLICIT. An exhaustive triangle-inequality scan identifies ten EXPLICIT matrices with structural violations (worst direct-edge ratio at least 1.2), which are outside every metric-dependent aggregate in this paper: `brg180`, `brazil58`, `gr120`, `gr48`, `gr24`, `gr21`, `bays29`, `hk48`, `dantzig42`, and `gr17`. That leaves 23 instances, seven of them EXPLICIT, and three of those (`swiss42`, `pa561`, and `fri26`) carry mild rounding-level violations, so conclusions are restricted to this screened set rather than to all non-Euclidean TSPs. We evaluate a hybrid pipeline that computes MST features from the original matrix and geometric features from a multidimensional-scaling (MDS) embedding. The paired reference is a constant multiple of the original-distance MST, a fixed $\alpha = 1.136$ close to the MAPE-minimizing constant 1.134 over the full 111-instance TSPLIB set. The MAPE-minimizing constant over these 23 instances is 1.1718 and reaches 7.55%, which is the floor any constant multiplier faces on this metric.

6.1 MDS Preprocessing

CEIL_2D cases use their native coordinates. For ATT, GEO, and EXPLICIT cases, classical MDS on the distance matrix produces an approximate positive-eigenvalue Euclidean representation [Torgerson, 1952, Gower, 1966, Borg and Groenen, 2005, Cox and Cox, 2000]. We select the smallest embedding dimension κ retaining at least 99.9% of positive eigenvalue mass, subject to $\kappa \leq 100$, and pass κ as the model's `dimension` feature. The symbol is distinct from the ascent budget k of the certified lower bound. The cap prevents five of 19 MDS cases from reaching 99.9%; hence this criterion does not imply distance preservation. Geometric features use the approximate embedding. MST features are computed from the original matrix D and are unaffected by embedding distortion. Appendix K reports the recovered distance diagnostics.

6.2 Results on Non-Euclidean TSPLIB95

On the instances it accepts, the hybrid pipeline clears the constant-multiplier floor. Table 9 reports the paired results by distance type.

GART 2.0 declines `si1032` rather than extrapolate below the greedy-to-MST ratio it was trained on. The dispersion gap is the substantive result: α ranges from 1.006 to 1.511 across the set, and a single multiplier cannot track that

Table 9: Paired evaluation on the 23 screened non-EUC_2D instances. GART 2.0 declines `si1032` at the coverage gate, so the Total row is not like-for-like.

Distance type	Count	Estimator	N	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	MSPE (%)
ATT	2	GART 2.0	2	1.42 [—, —]	3.66	3.66	-3.66
		Fixed $\alpha = 1.136$	2	4.12 [—, —]	3.38	3.38	-3.38
CEIL_2D	4	GART 2.0	4	5.16 [0.11, 6.35]	5.77	6.05	4.04
		Fixed $\alpha = 1.136$	4	6.44 [1.08, 7.86]	7.54	7.77	5.95
GEO	10	GART 2.0	10	2.60 [1.48, 3.23]	2.51	2.90	-1.63
		Fixed $\alpha = 1.136$	10	9.49 [2.13, 10.93]	9.62	4.10	-9.62
EXPLICIT screened	7	GART 2.0	6	3.11 [1.44, 3.87]	3.01	2.35	-2.15
		Fixed $\alpha = 1.136$	7	9.55 [5.04, 10.88]	8.12	10.16	1.75
Total	**23**	**GART 2.0**	**22**	**3.89 [2.52, 4.79]**	**3.34**	**3.21**	**-0.93**
		Fixed $\alpha = 1.136$	23	10.48 [6.94, 12.93]	8.26	6.29	-2.91

spread without trading one metric against the other. The advantage is not uniform. On the two ATT instances the fixed constant has the lower MAPE, 3.38% against 3.66%, and on CEIL_2D the two estimators lie within 1.3 percentage points of SDPE of one another. Groups of two and four instances cannot separate estimators. We do not estimate the relationship between per-instance embedding distortion and prediction error, so this remains a feasibility study for the hybrid pipeline rather than evidence of embedding robustness.

The Held–Karp relaxation never invokes the triangle inequality, so it is defined on any symmetric matrix and reads these instances directly. On the 29 instances both methods score over all 33 non-EUC_2D files — a set wider than the metric screen above, since the certificate tolerates non-metric input — GART 2.0 obtains 3.81% MAPE against 1.27% for the raw certified bound at an ascent budget of 100 under the Volgenant–Jonker step and 0.51% at a budget of 500 under the Polyak step, a factor of 7.5. The two steps are not interchangeable on this corpus, and Section 5.4 prices both. The bound also scores `brg180` and `si1032`, the two instances GART 2.0 declines. The MDS pipeline therefore extends a Euclidean estimator to these instances.

7 Conclusion

We present GART 2.0, a TSP tour-length estimator that predicts the MST-to-tour scaling factor α from features defined for arbitrary dimension. The features combine MST edge and topology statistics with centroid and coordinate-range descriptors, and no convex hull is computed. Across synthetic planar and multidimensional benchmarks and both the Euclidean and the screened non-Euclidean parts of TSPLIB95, it has the lowest aggregate MAPE and SDPE among the baselines scored on each of the four benchmark aggregates. It is ahead of a calibrated constant on the multidimensional set, of Çavdar–Sokol on planar i.i.d. uniform instances, and of the asymptotic ratio on TSPLIB EUC_2D, where the paired difference resolves and the dispersion gap is the wider of the two. On the multidimensional benchmark, dispersion falls monotonically across the size range and across the dimension range, including a dimension held out of training entirely. Where cost-level differences compress, the close-pair test of Section 4.7 still orders near-ties better than every baseline on all three benchmarks. The same trained model reads non-Euclidean distance matrices through an MDS embedding, well ahead of a fixed multiplier on the instances it accepts.

We then fix the boundary of the method against a certified Held–Karp 1-tree bound. The bound consumes no training corpus, proves its own output, and needs only a symmetric distance matrix. On the corpus aggregate the bound is the better instrument. It is more accurate on the multidimensional benchmark at lower cost, and more accurate again on the non-Euclidean instances it reads directly and GART 2.0 reaches only through an embedding. Above a few hundred nodes GART 2.0 is on the cost/accuracy front at every dimension it was trained on, and in the two larger size bands it is strictly better than the bound on both axes, from an ascent budget of 25 upward in the plane and over the mid-ladder budgets at $d = 3$. The estimator's advantage is scale. In the plane the measured cost exponents put the two families in different complexity classes, and an instance many times larger than anything in training is predicted in a fraction of a second. Below that size, off the training grid in dimension, or given only a distance matrix, the certified bound should be preferred.

The model weights, the scored benchmark outputs, and the scripts that regenerate every table and figure are released (Appendix A).

Accuracy depends on geometry far more than on scale. Error on the near-collinear class is several times what it is on the isotropic class. Section 4.6 diagnoses that failure as a training-coverage gap and a feature gap at once. Two pre-registered repair attempts were rejected and the released model is unchanged. Withholding a whole dimension or the large-n range from training costs a fraction of a MAPE point and appears as a systematic offset rather than as scatter. The estimator therefore degrades predictably off its training grid, and we demonstrate no correction for that offset. The labels are not uniformly certificates. Some are certified by a 1-tree bound that closes on the tour, some by exact dynamic programming, and the remainder carry an upper bound whose residual on the scored test split is narrow but not zero.

Both instruments in this paper are scored against reference tours in isolation. The open question is whether the precision advantage, or the certificate, changes the decisions a routing or partitioning algorithm makes when either is embedded inside it.

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A Code and Data Availability

Our implementation, trained model weights, and benchmark scripts are available at `https://github.com/sfeldmanMIG25/Area-and-Distribution-Free-Estimator-for-TSP`. Synthetic instances can be regenerated using the provided scripts with fixed random seeds. The per-class, per-dimension, per-size, rank-agreement, and paired-test grids are released as tidy tables alongside the per-instance predictions. The classical estimators and the calibrated constant-ratio baselines are released as separate modules, and the fitted constant-ratio table is frozen as a released artifact rather than refitted at inference. The benchmark results tables are generated from the scored outputs and written into the manuscript mechanically; a verification pass re-derives all 362 table cells and reports any that disagrees with the typeset value, and a second pass does the same for the 182 cells of the cost/accuracy ladders of Section 5. The remaining tables, and scalars quoted in the running prose such as the timing split and the quarantine and bound-check counts, are transcribed by hand from the same artifacts and are not covered by either check. The label validation of Appendix E, the refits of Section 4.5, the MDS diagnostics of Appendix K and the figures of this paper are each released as a standalone reproduction script, and the versions of the Python analysis dependencies are pinned in the repository. For direct audit of target clipping, the two released out-of-interval rows are `N1000_D15_G100_pchergphhtgncoe_59` (training, raw $\alpha = 0.974823$) and `N5_D2_G100_oc_33` (validation, raw $\alpha = 2.059518$).

The GART 2.0 pipeline uses SciPy [Virtanen et al., 2020] for Delaunay triangulation [Delaunay, 1934] and MST construction, Numba [Lam et al., 2015] for centroid-distance computation, NumPy [Harris et al., 2020] symmetric eigendecomposition for classical MDS, LightGBM [Ke et al., 2017] for regression, and scikit-learn [Pedregosa et al., 2011] for metric computation. Estimator timings were collected on one Intel i7 13th Gen machine with 64 GB RAM and Windows 11. Concorde 03.12.19 and LKH-3 3.0.13 generated the stored reference tour costs; the canonical table does not retain solver certificates.

B Training Data Generation Details

Synthetic TSP instances span four distribution classes, with the generator assigned independently per axis.

B.1 Training Corpus

The uniform class draws coordinates from $\mathcal{U}(0, G)$ where G is the grid size. The normal class draws from $\mathcal{N}(G/2, G/6)$ truncated to $[0, G]$. The clustered class places $k \in [3, 15]$ cluster centers uniformly and draws points from $\mathcal{N}(\mu_k, \sigma_k)$ around each center. The correlated class uses autoregressive structure across axes, $x_{j+1} = \rho \cdot x_j + \epsilon$ with j indexing the axis, producing axis-aligned anisotropy.

Node counts span $n \in [5, 1{,}000]$ with training-visible dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$, plus a held-out bucket at $d = 100$ (5,904 test instances). The pipeline ran Concorde and LKH-3 and retained the shorter tour cost.

The 106,272 solved instances are partitioned into training (69,768, 65.65%), validation (19,584, 18.43%), and test (16,920, 15.92%) using a stratified sample on $(d, n, \text{grid size})$. The stratification rule ranks instances within each stratum by a seeded uniform and assigns roughly the first tenth to test and the next fifth to validation, with the remainder to training; $d = 100$ instances bypass this rule and are locked to the test partition, which lifts the corpus-level test share to the 15.92% above. Table 11 lists the per-(d, n) training-instance count; grid size $G \in \{100, 1{,}000, 10{,}000\}$ is sampled uniformly within each cell.

B.2 2D Benchmark Generator Design

Four classes (isotropic, biased, geometric, clustered) follow conventions used in the TSP-instance-generation literature, including the variants reported by Çavdar and Sokol [2015] for the distribution-free baseline. The fifth class, Line

Noise, samples $y = mx + b + \epsilon$ with small ϵ and then clips out-of-range coordinates to the grid boundary rather than resampling them. Because the slope reaches ± 5 while x is drawn across the full grid width, most points fall outside the box and are pinned to a face. The realized geometry is therefore a union of two to four thin segments with dense collinear masses along the boundary, not a single noisy line: at $n \geq 200$ a median 59.9% of points lie exactly on a face, and α tracks that fraction closely (Spearman 0.86, rising monotonically from 1.180 to 1.431 across its quartiles) while showing no relationship to the transverse noise width. The `grid` member of the Geometric Struct. class draws a jittered square lattice, which no product of the four independent per-axis draws of Section 3.2 can produce, and the training corpus contains no lattice instance.

B.3 Multidimensional Benchmark Grid

The full (d, n) count grid of the 16,920-instance multidimensional benchmark:

Table 10: Multidimensional synthetic benchmark grid: instance count per (d, n) bucket (16,920 instances total).

$d \setminus n$	$[5, 10]$	$[11, 50]$	$[51, 100]$	$[101, 200]$	$[201, 500]$	$[501, 1000]$	Total
$d = 2$	162	108	135	27	81	135	**648**
$d = 3$–5	486	324	405	81	243	405	**1,944**
$d = 6$–10	810	540	675	135	405	675	**3,240**
$d = 15$–25	486	324	405	81	243	405	**1,944**
$d = 30$–50	810	540	675	135	405	675	**3,240**
$d = 100$	1,476	984	1,230	246	738	1,230	**5,904**
Total	**4,230**	**2,820**	**3,525**	**705**	**2,115**	**3,525**	**16,920**

Table 11: Training-set instance count per (d, n) cell (69,768 total).

$d \setminus n$	$[5, 10]$	$[11, 50]$	$[51, 100]$	$[101, 200]$	$[201, 500]$	$[501, 1000]$	Total
$d = 2$	1,026	684	855	171	513	855	**4,104**
$d = 3$–5	3,078	2,052	2,565	513	1,539	2,565	**12,312**
$d = 6$–10	5,130	3,420	4,275	855	2,565	4,275	**20,520**
$d = 15$–25	3,078	2,052	2,565	513	1,539	2,565	**12,312**
$d = 30$–50	5,130	3,420	4,275	855	2,565	4,275	**20,520**
Total	**17,442**	**11,628**	**14,535**	**2,907**	**8,721**	**14,535**	**69,768**

C Detailed Feature Specifications

Table 18 (Appendix I) lists the 11 geometric and centroid features, Tables 12 and 13 the 19 MST-derived ones, and Table 14 the single constructive ratio, 31 in all (Section 3.1). Listed complexities assume the MST has already been built; MST construction itself dominates the asymptotic cost and is analyzed in Section 3.

C.1 MST Edge Distribution Statistics

The moments are candidate predictors of the α ratio, not a tested causal mechanism.

Table 12: MST edge distribution statistics (10 features).

Feature Name	Description & Rationale	Complexity	Origin / Citation
MST Edge Max	Weight of the longest edge in the MST.	$O(n)$	Standard
MST Quantiles	10%, 25%, 50%, 75%, 90% percentiles. Provides a non-parametric distribution profile.	$O(n \log n)$	**Proposed**; via MST [Prim, 1957]
MST Edge Mean	Average edge length in the MST.	$O(n)$	Smith-Miles et al. [2010]
MST Edge Std	Standard deviation of edge lengths in the MST.	$O(n)$	Smith-Miles et al. [2010]
MST Edge Skewness	Asymmetry of edge distribution. High positive skew implies many short edges and few long bridges.	$O(n)$	Smith-Miles et al. [2010]
MST Edge Kurtosis	Tailedness of distribution. Indicates the presence of extreme outliers.	$O(n)$	Smith-Miles et al. [2010]

C.2 Topological Structure and Clustering Proxies

These features detect higher-order structures—clusters, filaments, hubs. We introduce the Dominance, Gap, and Normalized Diameter ratios as $O(n)$ candidate proxies for these properties. The Leaf Ratio rests on the hypothesis that low-leaf trees are more common on linear manifolds than in isotropic clouds. This interpretation is not evaluated independently here.

Table 13: Topological and clustering feature specifications (9 features).

Feature Name	Description & Rationale	Complexity	Origin / Citation
MST Dominance Ratio	Sum of largest $\sqrt{n}$ edges / Total Length. Captures weight of inter-cluster bridges.	$O(n)$	**Proposed**; via MST [Prim, 1957]
MST Gap Ratio	Max edge / Median edge. Measures magnitude of sparsity jumps.	$O(1)$	**Proposed**; via MST [Prim, 1957]
MST Leaf Ratio	Fraction of nodes with degree $k = 1$. Topology proxy for dimension.	$O(n)$	Smith-Miles et al. [2010]
MST Degree (Mean, Std)	Branching factor statistics.	$O(n)$	Smith-Miles et al. [2010]
MST Degree Max	Maximum node degree. Identifies "hub" nodes common in High-D spaces.	$O(n)$	Smith-Miles et al. [2010]
MST Diameter	Sum of edge weights on the longest path between any two leaves, computed via double-BFS.	$O(n)$	Graph Theory
MST Normalized Diameter	Diameter / Total Length. Scale-invariant shape proxy ($=$ 1.0 for path-manifolds exactly).	$O(1)$	**Proposed**; via double-BFS
Large Edge Count	Count of edges $> \mu + \sigma$. Secondary cluster indicator.	$O(n)$	**Proposed**; via MST [Prim, 1957]

C.3 Constructive Ratio

The third and last category holds a single feature. It is the only input GART 2.0 adds to the 30-feature block it inherits, it ranks second by SHAP magnitude (Appendix D), and it is the feature the non-Euclidean coverage gate of Section 6 tests.

D Feature Importance: SHAP Values

To characterize the 31-feature set, we report the SHAP values [Lundberg and Lee, 2017] against the released booster on a 5,000-row sample of the 16,920-row held-out test split. Table 15 ranks all 31 features by mean absolute SHAP contribution.

Table 14: Constructive ratio specification (1 feature).

Feature Name	Description & Rationale	Complexity	Origin / Citation
Greedy-to-MST Ratio	Length of an exact greedy nearest-neighbor tour divided by L_{MST}. A greedy tour is feasible, so the ratio is a constructed upper bound on the target α. Values outside $[1.035, 2.209]$, the range spanned by the full 106,272-row corpus, trigger the coverage gate; the training split alone spans $[1.0465, 2.1295]$.	$O(n^2d)$ dense for $n \leq 3000$; $O(nmd)$ with $m = 16$ candidate neighbors above it	**Proposed**; greedy nearest-neighbor construction

By group, the nineteen MST-derived features carry 50.7% of total mean absolute SHAP magnitude, the eleven geometric features 25.5%, and the single constructive ratio 23.9%. SHAP magnitude describes model reliance, not a causal effect or proof that a feature group improves generalization.

SHAP analysis served as a diagnostic during feature reduction, yielding the 31-feature set described in Section 3.1; because this selection used validation feedback, it should not be interpreted as an independent confirmatory test.

Table 15: All 31 features ranked by mean absolute SHAP share, against the released booster on a 5,000-row test sample.

Rank	Feature	Mean $\lvert$SHAP$\rvert$	Share (%)
1	`mst_dominance_ratio`	0.2866	26.52
2	`greedy_nn_over_mst`	0.2578	23.86
3	`n_customers`	0.0922	8.53
4	`mst_diameter_normalized`	0.0886	8.20
5	`dimension`	0.0605	5.60
6	`log_bounding_hypervolume`	0.0513	4.75
7	`mst_degree_mean`	0.0414	3.84
8	`bounding_hypervolume`	0.0261	2.41
9	`mst_gap_ratio`	0.0195	1.81
10	`centroid_dist_std`	0.0193	1.79
11	`large_edge_count`	0.0183	1.70
12	`mst_leaf_ratio`	0.0179	1.66
13	`mst_diameter`	0.0154	1.42
14	`mst_degree_std`	0.0119	1.10
15	`mst_edge_skew`	0.0069	0.64
16	`aspect_ratio`	0.0068	0.63
17	`mst_degree_max`	0.0063	0.59
18	`centroid_dist_mean`	0.0059	0.55
19	`mst_edge_std`	0.0051	0.47
20	`centroid_dist_iqr`	0.0046	0.42
21	`mst_edge_q25`	0.0045	0.42
22	`mst_edge_max`	0.0045	0.42
23	`mst_edge_kurtosis`	0.0044	0.41
24	`mst_edge_q75`	0.0039	0.36
25	`node_density`	0.0035	0.33
26	`mst_edge_q90`	0.0035	0.33
27	`mst_edge_mean`	0.0029	0.27
28	`mst_edge_q50`	0.0029	0.27
29	`mst_edge_q10`	0.0027	0.25
30	`centroid_dist_max`	0.0026	0.24
31	`log_node_density`	0.0026	0.24

E Label Validation and Certification

Every label is screened before any table is built. We recompute the length of every stored tour on its own released coordinates, over all $106{,}272$ instances; a stored cost that its own tour cannot reproduce to within rounding describes no released point set, and no bound is tight enough to reconstruct the optimum from the coordinates alone. The 184 instances (0.173%) that fail are *quarantined* rather than scored: they carry an empty reference cost and the status `quarantined_label_unrecoverable` in every released results CSV, an aggregate that ignores the status column yields NaN and never a plausible number, and the scoring harness raises on any attempt to score one. 74 of the 184 fall in the multidimensional test partition, which is therefore scored on the remaining $16{,}846$ instances throughout.

A certified lower bound is also an acceptance test. If B is a proven lower bound on an instance's optimum and L is its stored label, then $B \leq \mathrm{OPT} \leq L$ must hold whenever L is an upper bound on the optimum, and $B > L$ refutes the label outright, with no tour entering the test. Every reference cost is recomputed in float64 from the released coordinates: at $n \leq 10$ the label is the exact Held–Karp dynamic-programming optimum, a certificate; above that size the label is the float64 length of the retained tour, an upper bound that is exact wherever that tour is optimal; and where a 1-tree bound closes on the tour the label is certified after the fact, 2,182 instances. The 2D benchmark's labels are the better of the stored tour and a float-metric 2-opt descent from it, because a solver that minimizes a rounded metric can select a tour the float64 metric can beat. TSPLIB is scored on its published optima, with one documented exception: `linhp318` declares a fixed-edge Hamiltonian-path problem on `lin318`'s coordinates, so its published 41,345 is a path optimum no tour on those coordinates can attain; every estimator and the bound are scored there against the tour optimum on those coordinates, 42,029, which is `lin318`'s published value; a scan of the full instance set found no second fixed-edge declaration and no second duplicated geometry.

Across the 145,013 bound checks the released artifacts supply there are zero violations of $B \leq L$, the worst relative excess being 5.8×10^{-15}. The 184 quarantined instances above carry no recoverable label and are outside it. And seven TSPLIB instances carry no bound, because the test runs in TSPLIB's integer metric there and the dense perturbed matrix does not fit in memory, five of them inside the scored EUC_2D set; nothing in this section is claimed about those instances.

F Benchmark-Model Implementation Notes

BHH. For points drawn i.i.d. from a density f of bounded support, $L_n/n^{(d-1)/d} \to \beta_d \int f(x)^{(d-1)/d}\,dx$ [Beardwood et al., 1959]; for f uniform on a region of measure $\mathcal{V}$ the integral is $\mathcal{V}^{1/d}$, giving $\beta_d n^{(d-1)/d}\mathcal{V}^{1/d}$. Here $\mathcal{V}$ is the measure of the sampling region, and the convex hull of the realized sample is a different, downward-biased statistic that converges to it only as n grows. Our synthetic generators draw on $[0,G]^d$, so $\mathcal{V} = G^d$ is known exactly, and that form is reported on the matched domain. We use $\beta_2 = 0.7124 \pm 0.0002$ [Johnson et al., 1996] and $\beta_3 = 0.6979 \pm 0.0002$ [Percus and Martin, 1996]. For $d \geq 4$ no comparably pinned estimate exists and we substitute the large-d asymptotic $\sqrt{d/(2\pi e)}$ [Bertsimas and van Ryzin, 1990], a form that source proves for the minimum-spanning-tree and minimum-matching constants and only conjectures for the TSP constant; that substitution, and the fact that $n \leq 1000$ with $d \geq 4$ is far from the asymptotic regime, both bound how much the $d \geq 4$ rows can be read as a test of the theorem.

Not scored: Daganzo, Chien, and Kwon–Golden–Wasil. We survey all three in Section 1.2 and score none of them. Their equations reach the modern record through the literature review in Figliozzi [2009], not through the articles themselves: Daganzo [1984a], which is the Daganzo paper that review carries, Chien [1992] and Kwon et al. [1995] are paywalled, a direct check of each DOI returned no open-access location. Nor is the secondary record self-consistent. For Chien, Çavdar and Sokol [2015] give a Daganzo-form coefficient of 0.69 and Choi [2021] give 0.88, against the 0.67 of the $2.1\bar{r} + 0.67\sqrt{nR}$ form that review carries; for Kwon–Golden–Wasil, the published renderings disagree on whether the bracket carries n or $(n{+}1)$ and on the precision of its coefficients. Choosing between those renderings requires the primaries, so no number is printed for any of the three. The scored pair, BHH and Çavdar–Sokol, are the two whose primary documents are held.

Çavdar–Sokol. This estimator is implemented directly from a primary document, Çavdar [2014], Chapter 4, the same work as Çavdar and Sokol [2015] and by the same author. The area term is the area of the rectangle covering the nodes, and $\bar{c}_j$ and cstdev_j are the mean and standard deviation of the absolute distances to that rectangle's midpoint line on axis j, not to the sample mean. The divisor $\bar{c}_x\bar{c}_y$ sits inside the radical; placing it outside leaves the second term dimensionless. Two further elements of the published method follow. The first is the frame. The estimator is defined on an axis-aligned rectangle and is not rotation invariant, so the implementation rotates each instance into its minimum-area enclosing rectangle by scanning convex-hull edges [Freeman and Shapira, 1975], as the source prescribes. The second is the finite-n correction $0.9325e^{0.00005298n} - 0.2972e^{-0.01452n}$, Eq. (21) of the dissertation, fitted over $n \in \{100, 125, \ldots, 975\}$ at $R^2 = 0.9867$; the raw estimate is divided by it. Choi [2021] records that the uncorrected estimator underestimates for $n < 1000$, which is the direction the correction moves it. That correction is bounded to the range it was fitted over, and the bound binds on this benchmark: 1,320 of the 2,580 2D instances have $n < 100$, so for slightly over half the benchmark the ratio is evaluated at the $n = 100$ endpoint rather than extrapolated below it. The authors' own figures show the ratio continuing to fall toward 0.4 as n approaches 10, but publish no fitted form there, so extrapolating would substitute our curve for theirs. We hold the upper end fixed for the same reason: above $n = 975$ no correction is applied, because Eq. (21) grows without bound ($E/T = 1.22$ at $n = 5000$) and would contradict the training fit it exists to repair. That leaves a 1.8% step at the upper boundary, where the fitted ratio reads 0.982. Two properties of the source, not of our evaluation, remain: the regression targets Lin–Kernighan tour lengths rather than optima, and every training graph carried a node at each corner of its rectangle, which pins the graph area to the covering rectangle in a way our instances do not.

Constant-ratio references. The $\alpha = 1$ row returns L_{MST}. The asymptotic row uses $\beta_{\mathrm{TSP}}/\beta_{\mathrm{MST}}$ for uniform planar points. The numerator is well established at 0.7124; the denominator is not. We quote 0.6331 following Cortina-Borja and Robinson [2000]. That figure is a simulation estimate and not a theorem, and published estimates of the planar Euclidean MST constant span a range wider than its printed precision, so the ratio 1.1253 should be read as approximate. The calibrated rows take the mean of α within each dimension, and within each dimension–size cell, over the training split only; the fitted values are frozen and released rather than refitted at inference. At $d = 100$, which no training row covers, the table extrapolates from $d = 50$.

Custom Hilbert sort. We independently min–max normalize every axis to a 16-bit grid, sort by a d-dimensional Hilbert index [Hilbert, 1891], visit nodes in that order, and close the tour. This is a custom generalization inspired by the planar space-filling-curve heuristic of Bartholdi and Platzman [1982]; its planar performance guarantee does not transfer to our anisotropically normalized multidimensional ordering. It constructs a tour rather than estimating one, so its error is bounded below by the gap between a space-filling-curve tour and the optimum.

Hybrid non-Euclidean builder. The hybrid builder of Section 6 computes every feature the released model consumes. Bounding hypervolume is stored as $\exp(\min(\log \mathrm{hv},\ 690))$, so the raw feature stays finite at high embedding dimension while its log-scale companions carry the magnitude past the cap.

Convex-hull cap. Where an estimator receives the convex-hull plug-in we compute the hull only for $d \leq 3$ and substitute the coordinate bounding box above it, recording which was used. QHull's output size grows as $n^{\lfloor d/2 \rfloor}$, the same blow-up cited in Section 1 as a reason not to build estimators on hull geometry; at $d = 8$ with $n = 1000$ a single hull dominates the entire benchmark's runtime.

G Classical Estimators: Full Benchmark and Matched Domain

Table 16 reports the two classical estimators twice. The upper panels apply them to the complete 2D and TSPLIB EUC_2D benchmarks, with the convex hull of each instance standing in for BHH's region measure, which is the only option on TSPLIB and the fairer option on the degenerate 2D classes. The lower panel restricts to the 210 i.i.d. uniform 2D instances, where the sampling region $[0, G]^2$ is exact and BHH receives it. Çavdar–Sokol receives the same input in both, the rectangle covering the nodes, so only its instance set changes between the panels.

Table 16: Classical estimators on the full benchmarks and on their matched domains; a large MSPE relative to MAPE identifies a systematic offset.

Domain	Estimator	N	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	MSPE (%)
Full 2D benchmark	BHH	2,580	32.00 [30.00, 33.97]	23.80	15.53	-13.71
	Çavdar–Sokol	2,580	34.61 [31.96, 37.12]	18.24	6.63	-0.96
	GART 2.0	2,580	4.68 [4.42, 4.92]	2.89	1.49	-0.10
	Calibrated MST ratio $\hat{\rho}(d, n)$	2,580	9.38 [8.96, 9.78]	5.76	2.31	-0.75
TSPLIB EUC_2D	BHH	78	40.75 [26.25, 53.61]	25.14	13.41	13.66
	Çavdar–Sokol	78	36.34 [24.38, 48.39]	23.87	11.32	20.07
	GART 2.0	78	2.92 [2.40, 3.38]	2.53	1.87	2.08
	Calibrated MST ratio $\hat{\rho}(d, n)$	78	4.48 [3.59, 5.32]	3.74	3.18	1.18
2D `random` (uniform on $[0, G]^2$) matched domain	**GART 2.0**	210	2.07 [1.65, 2.48]	1.31	0.75	-0.01
	GART 1.0	210	4.96 [4.53, 5.35]	5.75	6.05	4.69
	BHH (sampling region)	210	7.76 [6.39, 8.98]	8.91	6.70	-8.65
	Çavdar–Sokol	210	14.28 [11.63, 16.53]	8.16	2.84	-5.97
	L_{MST} ($\alpha = 1$)	210	7.70 [6.39, 8.82]	15.60	12.52	-15.60

H Per-Size Detail for the TSPLIB95 EUC_2D Benchmark

Table 17: TSPLIB95 EUC_2D benchmark. Time is median wall clock; within each bucket rows are ordered by SDPE.

n range	Estimator	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	MSPE (%)	R^2_α	Time (ms)
$n \in [51, 150]$ Count=23	**GART 2.0**	**2.63** [1.34, 3.58]	2.01	1.22	1.47	0.580	3.64
	L_{MST} ($\alpha = 1$)	3.77 [2.51, 4.50]	13.43	12.67	-13.43	-9.539	1.68
	Asymptotic MST ratio	4.24 [2.82, 5.07]	3.55	2.13	-2.58	-0.395	1.16
	Calibrated MST ratio $\hat{\rho}(d, n)$	4.68 [3.07, 5.68]	4.09	3.93	1.04	-0.176	1.14
	GART 1.0	5.64 [3.58, 7.19]	5.44	3.92	4.07	-1.307	4.05
	Custom Hilbert sort	11.46 [8.20, 13.91]	39.91	38.92	39.91	—	0.656

Table 17 continued from the previous page

n range	Estimator	SDPE (%) [95% CI]	MAPE (%)	|Median| (%)	MSPE (%)	R^2_α	Time (ms)
$n \in [151, 400]$ Count=16	**GART 2.0**	**2.86** [1.95, 3.51]	2.38	2.50	1.51	0.706	4.51
	L_{MST} ($\alpha = 1$)	4.82 [2.13, 6.65]	12.38	11.56	-12.38	-4.801	2.33
	Calibrated MST ratio $\hat{\rho}(d, n)$	5.37 [2.49, 7.39]	3.73	2.43	0.01	0.049	1.91
	Asymptotic MST ratio	5.42 [2.39, 7.48]	3.39	1.27	-1.41	-0.088	1.84
	GART 1.0	5.53 [3.93, 6.54]	5.92	4.46	4.93	-0.497	5.03
	Custom Hilbert sort	7.35 [3.56, 10.39]	44.24	42.42	44.24	—	1.44
$n > 400$ Count=39	**GART 2.0**	**3.05** [2.32, 3.62]	2.90	1.98	2.68	-0.071	20.5
	L_{MST} ($\alpha = 1$)	3.58 [2.84, 4.17]	9.74	9.39	-9.74	-6.448	9.55
	Calibrated MST ratio $\hat{\rho}(d, n)$	3.95 [3.13, 4.60]	3.54	3.24	1.75	-0.122	9.52
	Asymptotic MST ratio	4.03 [3.20, 4.70]	3.57	3.15	1.57	-0.132	9.27
	GART 1.0	7.64 [6.07, 8.92]	11.23	9.66	11.18	-10.726	18.4
	Custom Hilbert sort	12.55 [9.60, 15.71]	48.66	47.03	48.66	—	9.14
Total Count=78	**GART 2.0**	**2.92** [2.40, 3.38]	2.53	1.87	2.08	0.484	6.12
	L_{MST} ($\alpha = 1$)	4.21 [3.30, 4.99]	11.37	11.56	-11.37	-5.591	3.08
	Calibrated MST ratio $\hat{\rho}(d, n)$	4.48 [3.59, 5.32]	3.74	3.18	1.18	0.087	2.62
	Asymptotic MST ratio	4.74 [3.72, 5.62]	3.53	2.67	-0.27	-0.010	2.57
	GART 1.0	7.45 [6.24, 8.46]	8.43	6.51	7.80	-3.579	6.12
	Custom Hilbert sort	11.85 [9.86, 13.59]	45.17	44.60	45.17	—	2.64

I Geometric and Centroid Feature Specifications

Table 18: Geometric and centroid feature specifications (11 features).

Feature Name	Description & Rationale	Complexity	Origin / Citation
Dimension (d)	The number of spatial coordinates defining each point.	$O(1)$	Standard
Number of Nodes (n)	The total count of nodes in the instance; released feature name `n_customers`.	$O(1)$	Standard
Aspect Ratio	Ratio of the largest dimension range to the smallest. Detects subspace distortion.	$O(n \cdot d)$	**Proposed**
Bounding Hypervolume	Product of coordinate ranges: $\prod_{j=1}^{d}(\max_j - \min_j)$. Approximates the sampling-region measure $\mathcal{V}$ for BHH bounds; stored capped at e^{690}, beyond which the log twin carries the scale.	$O(n \cdot d)$	Adapted from Beardwood et al. [1959]
Log Bounding Hypervolume	Logarithm of hypervolume: $\sum_{j=1}^{d} \ln(\max_j - \min_j)$. Log-space stable for $d \geq 100$.	$O(n \cdot d)$	**Proposed**
Node Density	Ratio of n to Bounding Hypervolume. Proxy for point intensity $\delta = n/\mathcal{V}$.	$O(1)$	**Proposed**; cf. Beardwood et al. [1959]
Log Node Density	Logarithm of node density: $\ln(n) - \sum \ln(\max_d - \min_d)$. Log-space stable.	$O(1)$	**Proposed**
Centroid Dist (Mean, Std)	Average and deviation of distances to the centroid. Measures central tendency.	$O(n \cdot d)$	Adapted from Çavdar and Sokol [2015]
Centroid Dist Max	Radius of the enclosing sphere.	$O(n \cdot d)$	Adapted from Çavdar and Sokol [2015]
Centroid Dist IQR	Interquartile range of centroid distances. Robust dispersion metric.	$O(n \cdot d)$	**Proposed**; extends Çavdar and Sokol [2015]

J Trained Hyperparameters and Booster Statistics

Table 19 reports the hyperparameters and the resulting booster statistics for the final GART 2.0 model. The search we rejected was Optuna's tree-structured Parzen estimator (TPE), 200 trials, 60 of them completed and 140 pruned, against in-distribution validation. The tuned booster reaches 0.6112% on the multidimensional test split against 0.6226% for the same booster on the frozen hyperparameters, and 2.6364% against 2.4725% on TSPLIB EUC_2D; both are unconstrained and both are on the logit target, so the monotone constraint of the released model does not enter the difference, and scoring the tuned booster against the released model instead would confound the search with that constraint.

Table 19: GART 2.0 hyperparameters and trained-model statistics; the hyperparameter block is inherited and frozen.

Group	Parameter	Value
Hyperparameters (inherited, frozen, not tuned)		
Tree structure	num leaves	148
	min child samples	23
	feature fraction	0.548
	bagging fraction / freq	0.609 / 6
Optim. & reg.	learning rate	0.0260
	L1 regularization	7.76×10^{-2}
	L2 regularization	1.19×10^{-5}
Training configuration		
Training setup	boosting type	gbdt
	objective	regression_l2 (squared error)
	early-stopping patience	100
	random seed	42
Trained-booster statistics		
Booster	trees (after early stop)	1,118
	avg leaves / tree	147.80
	max leaves / tree	148
	avg tree depth	11.83
	max tree depth	40

Two other learners were fitted on the same feature pipeline before the gradient-boosted ensemble was frozen, and neither displaced it. A residual feed-forward network predicts the same α from the same MST-and-geometry descriptors, and on the 2D diverse benchmark it costs 137.5 to 146.2 ms per instance against GART 2.0's 103.3 to 147.8 ms, slower on five of the six generator classes and slower in aggregate. It does not buy accuracy with that time so much as move it: the network reads 2.98% MAPE overall against 2.89% and loses on five of the six classes, but it wins the near-collinear Line Noise class at 9.14% against 10.82% and finishes marginally ahead on aggregate dispersion, 4.61% SDPE against 4.68%. An ordinary least-squares fit is the cheapest of the three at 61.9 to 92.1 ms and pays for it in the tail, reaching 17.9% MAPE on the clustered class against 2.2%, eight times the released model's error, at $R^2_\alpha = -9.53$. Neither is a control on the released feature vector and we do not report them as one: the network was fitted on 30 columns and the linear model on 28 against the released block's 31, and neither was fitted on the released feature table.

K MDS Embedding for Non-Euclidean Instances

For TSPLIB ATT, GEO (geographic coordinates with great-circle distances), and EXPLICIT (distance matrix only) instances, classical MDS gives an approximate positive-eigenvalue Euclidean representation. CEIL_2D uses native coordinates. The embedding dimension κ is chosen by the rule of Section 6; this retained-mass rule is a dimension-selection heuristic, not a distance-preservation guarantee.

Given an $n \times n$ distance matrix D, classical MDS computes the eigendecomposition of the doubly-centered matrix $B = -\frac{1}{2}JD^{(2)}J$ where $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ and $D^{(2)}$ is the element-wise squared distance matrix. The embedding coordinates are $X = V_\kappa \Lambda_\kappa^{1/2}$ where V_κ and Λ_κ are the top κ eigenvectors and eigenvalues.

We embed into κ dimensions, selecting the smallest κ that retains $\geq$99.9% of positive eigenvalue mass (capped at $\kappa = 100$). For GEO instances, the selected dimension ranges from 2 to 61; for screened EXPLICIT instances, from 10 to 100. The MST features (edge weights, degrees, diameter) are computed from the original distance matrix, while geometric features (centroid statistics, bounding volume, aspect ratio) use the approximate MDS coordinates. The dimension κ is passed to the model as the `dimension` feature.

Normalized distance stress is $\sqrt{\sum_{i<j}(D_{ij} - \hat{D}_{ij})^2 / \sum_{i<j} D_{ij}^2}$, where $\hat{D}_{ij}$ is the Euclidean distance induced by the retained MDS coordinates; correlation is Pearson correlation between the same distance vectors. Table 20 reports medians and ranges. CEIL_2D stress measures only integer-ceiling distortion on a fixed-seed sample of 100,000–1,000,000 pairs; other rows use all pairs. Five capped instances do not reach the 99.9% target: `att532`, `pa561`, `si175`, `si535`, and `si1032`.

Table 20: Embedding diagnostics on the 23 screened instances; medians with [minimum, maximum]. "Below target": eigenvalue mass under 99.9% from the $\kappa = 100$ cap.

Distance type	N	κ range	Normalized stress	Distance correlation	Below target
CEIL_2D	4	2	0.0000 [0.0000, 0.0000]	1.00000 [1.00000, 1.00000]	0
ATT	2	6–100	0.0028 [0.0007, 0.0048]	0.99999 [0.99998, 1.00000]	1
GEO	10	2–61	0.0207 [0.0005, 0.2363]	0.99951 [0.96944, 1.00000]	0
EXPLICIT screened	7	10–100	0.0198 [0.0041, 0.1208]	0.99954 [0.99598, 0.99999]	4

The screened application set excludes ten EXPLICIT matrices with structural triangle-inequality violations. Seven EXPLICIT matrices remain, including three with mild rounding-level violations. GART 2.0 declines one of them, `si1032`, and obtains 3.01% MAPE on the six it accepts. Distortion varies substantially within type (maximum stress 0.2363 for GEO and 0.1208 for screened EXPLICIT), so we do not attribute between-type accuracy differences to the distance label alone.